Index

Sr. No.	Name of the Practicals	Date	Sign.
1	**Introduction to Excel** • Perform conditional formatting on a dataset using various criteria. • Create a pivot table to analyze and summarize data. • Use VLOOKUP function to retrieve information from a different worksheet or table. • Perform what-if analysis using Goal Seek to determine input values for desired output.	**15/11/2025**	
2	**Data Frames and Basic Data Pre-processing** • Read data from CSV and JSON files into a data frame. • Perform basic data pre-processing tasks such as handling missing values and outliers. • Manipulate and transform data using functions like filtering, sorting, and grouping.	**22/11/2025**	
3	**Feature Scaling and Dummification** • Apply feature-scaling techniques like standardization and normalization to numerical features. • Perform feature dummification to convert categorical variables into numerical representations.	**29/11/2025**	
4	**Hypothesis Testing** • Formulate null and alternative hypotheses for a given problem. • Conduct a hypothesis test using appropriate statistical tests (e.g., t-test, chi- square test). • Interpret the results and draw conclusions based on the test outcomes.	**06/12/2025**	
5	**ANOVA (Analysis of Variance)** • Perform one-way ANOVA to compare means across multiple groups. • Conduct post-hoc tests to identify significant differences between group means.	**20/12/2025**	
6	**Regression and Its Types** • Implement simple linear regression using a dataset. • Explore and interpret the regression model coefficients and goodness-of-fit measures. • Extend the analysis to multiple linear regression and assess the impact of additional predictors.	**03/01/2026**	

Sr. No.	Name of the Practicals	Date	Sign.
7	**Logistic Regression and Decision Tree** • Build a logistic regression model to predict a binary outcome. • Evaluate the model's performance using classification metrics (e.g., accuracy, precision, recall). • Construct a decision tree model and interpret the decision rules for classification.	**10/01/2026**	
8	**K-Means Clustering** • Apply the K-Means algorithm to group similar data points into clusters. • Determine the optimal number of clusters using elbow method or silhouette analysis. • Visualize the clustering results and analyze the cluster characteristics.	**24/01/2026**	
9	**Principal Component Analysis (PCA)** • Perform PCA on a dataset to reduce dimensionality. • Evaluate the explained variance and select the appropriate number of principal components. • Visualize the data in the reduced-dimensional space.	**07/02/2026**	
10	**Data Visualization and Storytelling** • Create meaningful visualizations using data visualization tools. • Combine multiple visualizations to tell a compelling data story. • Present the findings and insights in a clear and concise manner.	**14/02/2026**	

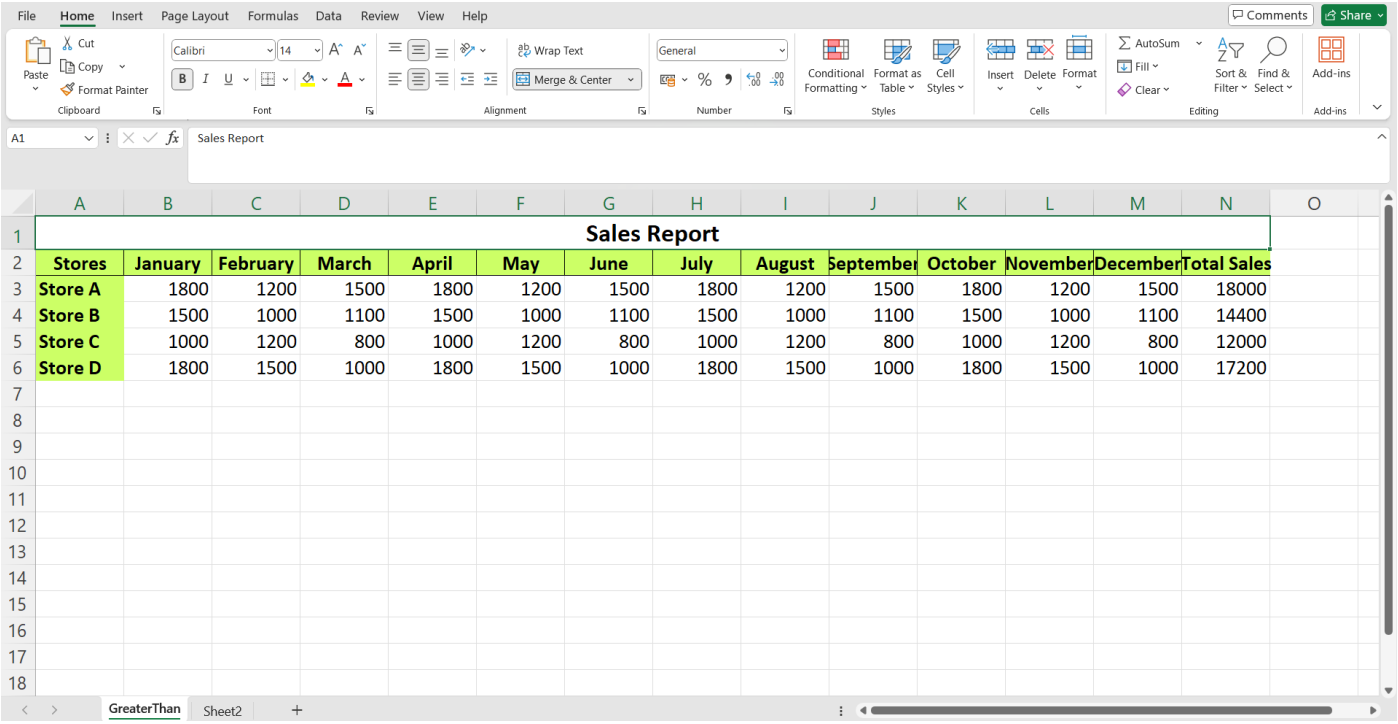
Practical 1

Aim: Introduction to Excel

- Perform conditional formatting on a dataset using various criteria.
- Create a pivot table to analyze and summarize data.
- Use VLOOKUP function to retrieve information from a different worksheet or table.
- Perform what-if analysis using Goal Seek to determine input values for desired output.

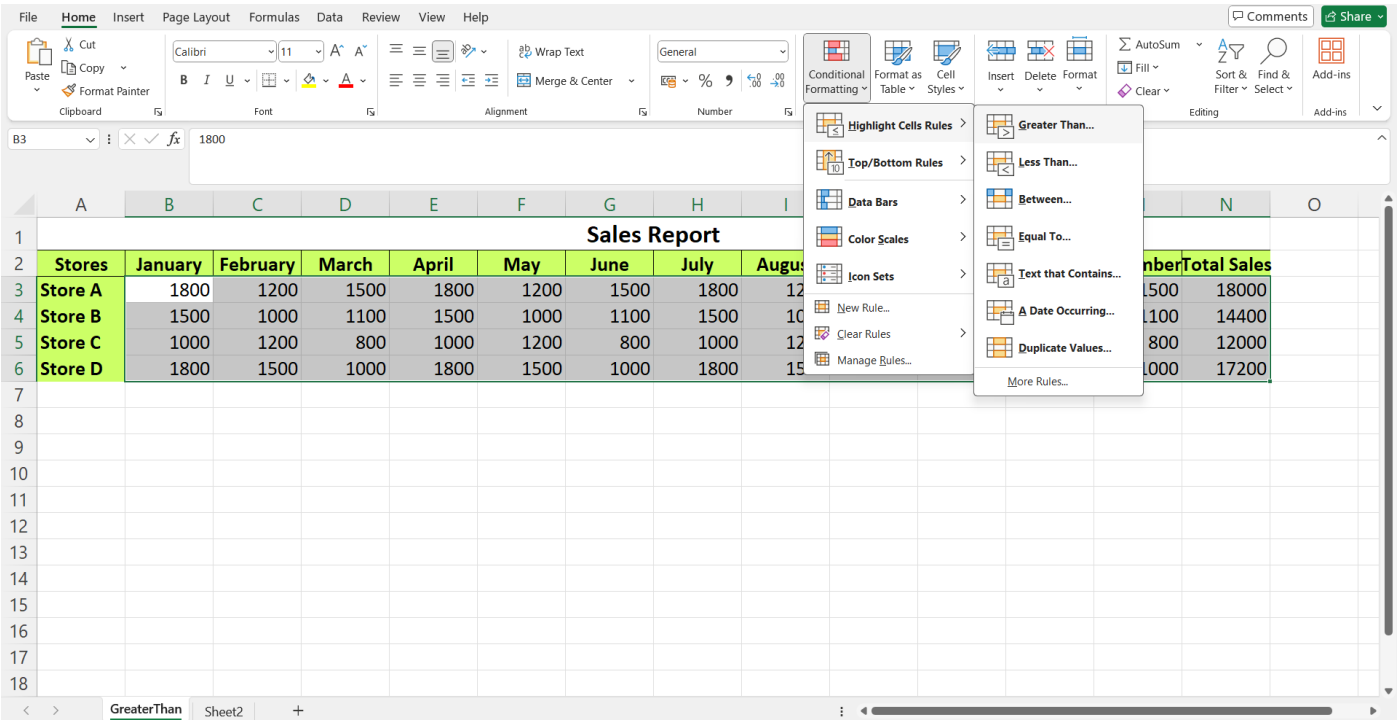
Steps

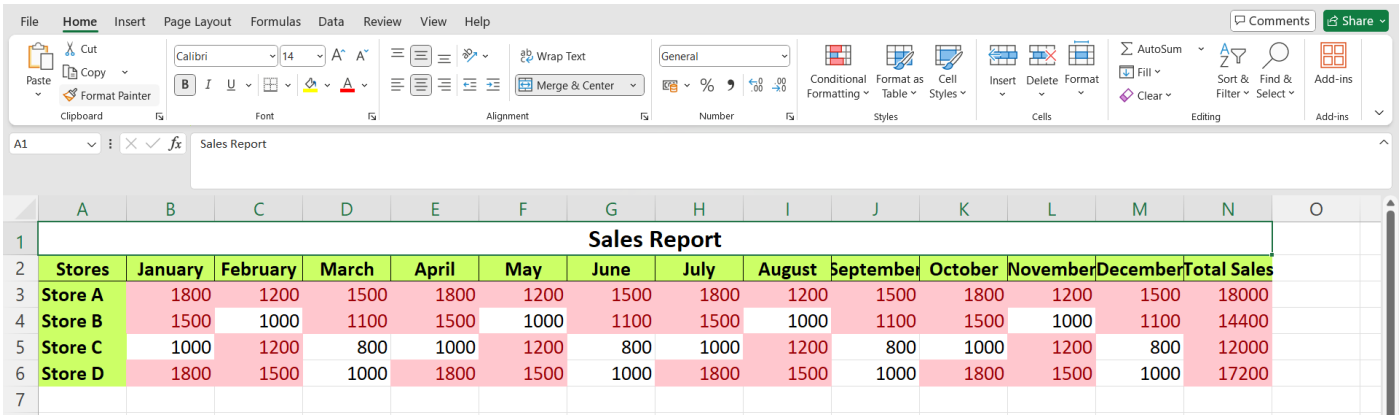
Step 1 : Go to conditional formatting > Greater Than



Sales Report													
Stores	January	February	March	April	May	June	July	August	September	October	November	December	Total Sales
Store A	1800	1200	1500	1800	1200	1500	1800	1200	1500	1800	1200	1500	18000
Store B	1500	1000	1100	1500	1000	1100	1500	1000	1100	1500	1000	1100	14400
Store C	1000	1200	800	1000	1200	800	1000	1200	800	1000	1200	800	12000
Store D	1800	1500	1000	1800	1500	1000	1800	1500	1000	1800	1500	1000	17200

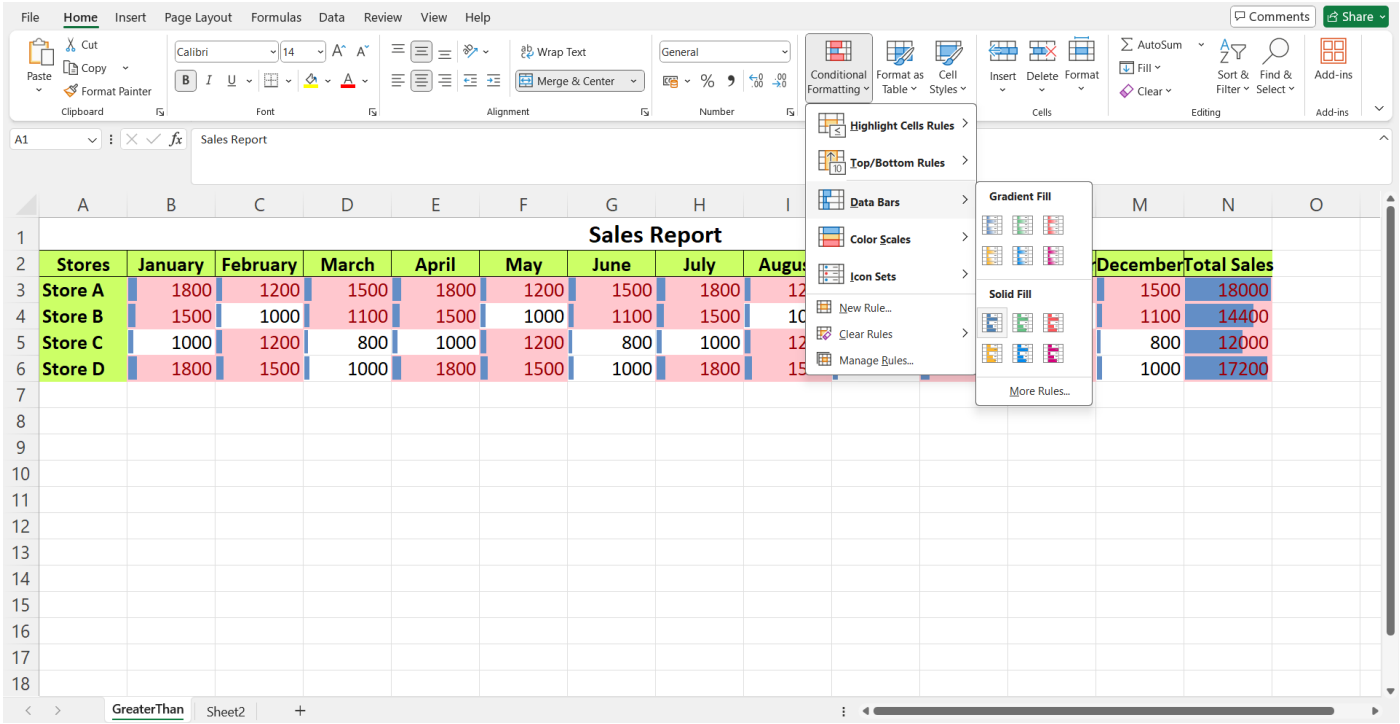
Step 2: Enter the greater than filter value for example 1000.

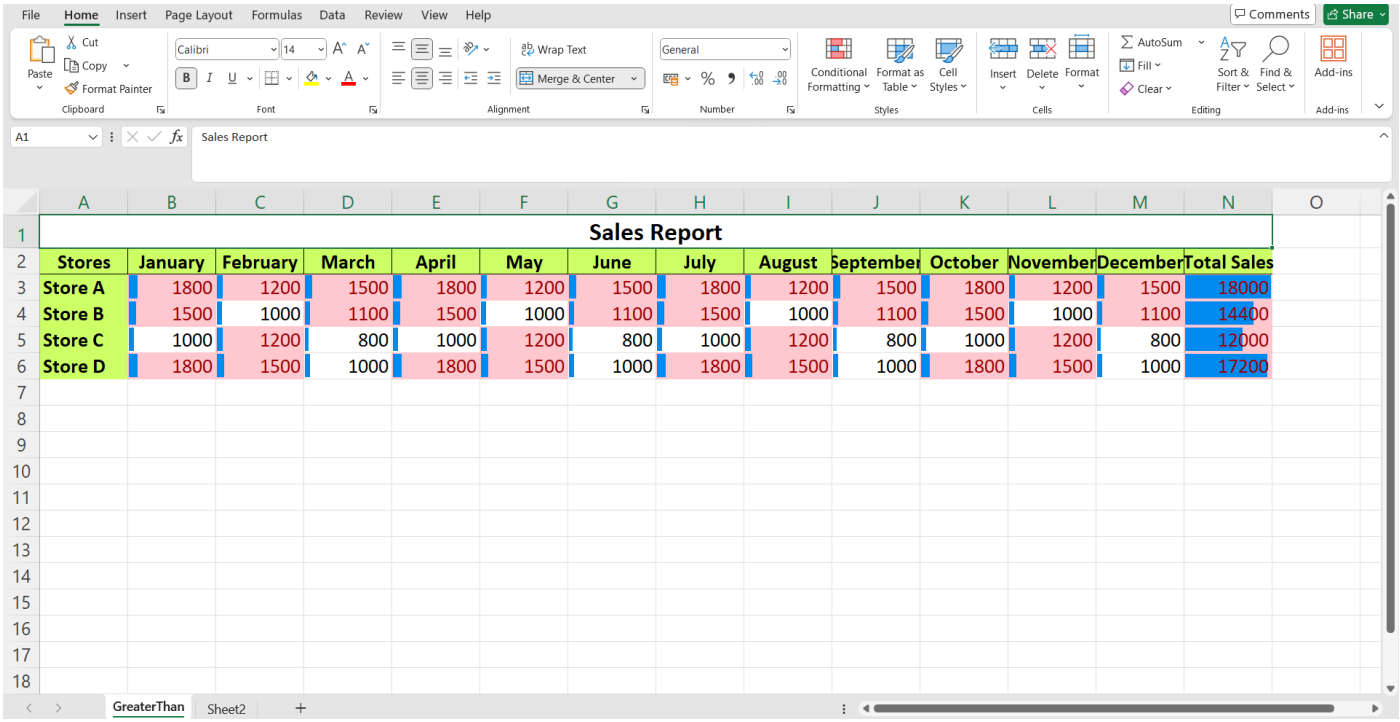




Sales Report													
Stores	January	February	March	April	May	June	July	August	September	October	November	December	Total Sales
Store A	1800	1200	1500	1800	1200	1500	1800	1200	1500	1800	1200	1500	18000
Store B	1500	1000	1100	1500	1000	1100	1500	1000	1100	1500	1000	1100	14400
Store C	1000	1200	800	1000	1200	800	1000	1200	800	1000	1200	800	12000
Store D	1800	1500	1000	1800	1500	1000	1800	1500	1000	1800	1500	1000	17200

Step 3: Go to Data Bars > Solid Fill in conditional formatting.

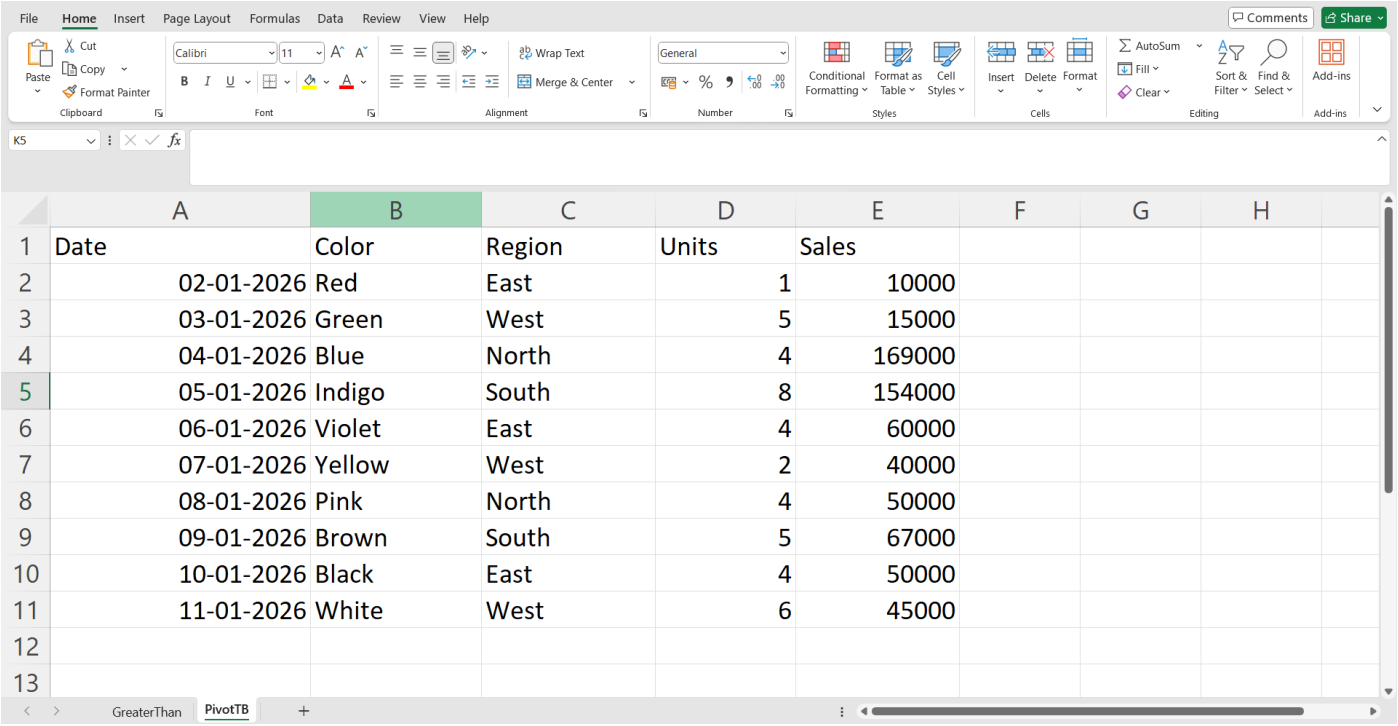




Pivot Table

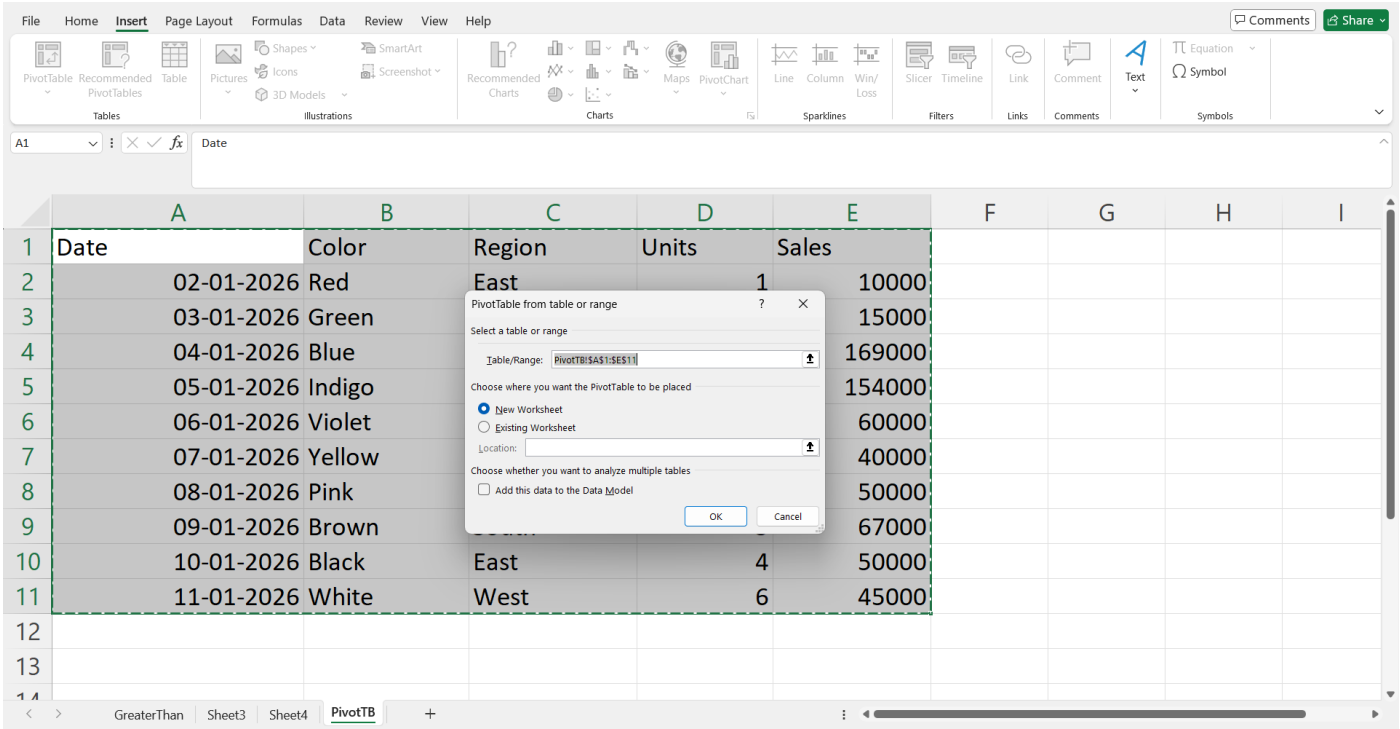
Steps:-

Step 1: First Create the Excel Data and select the entire table and go to Insert tab PivotChart > Pivotchart

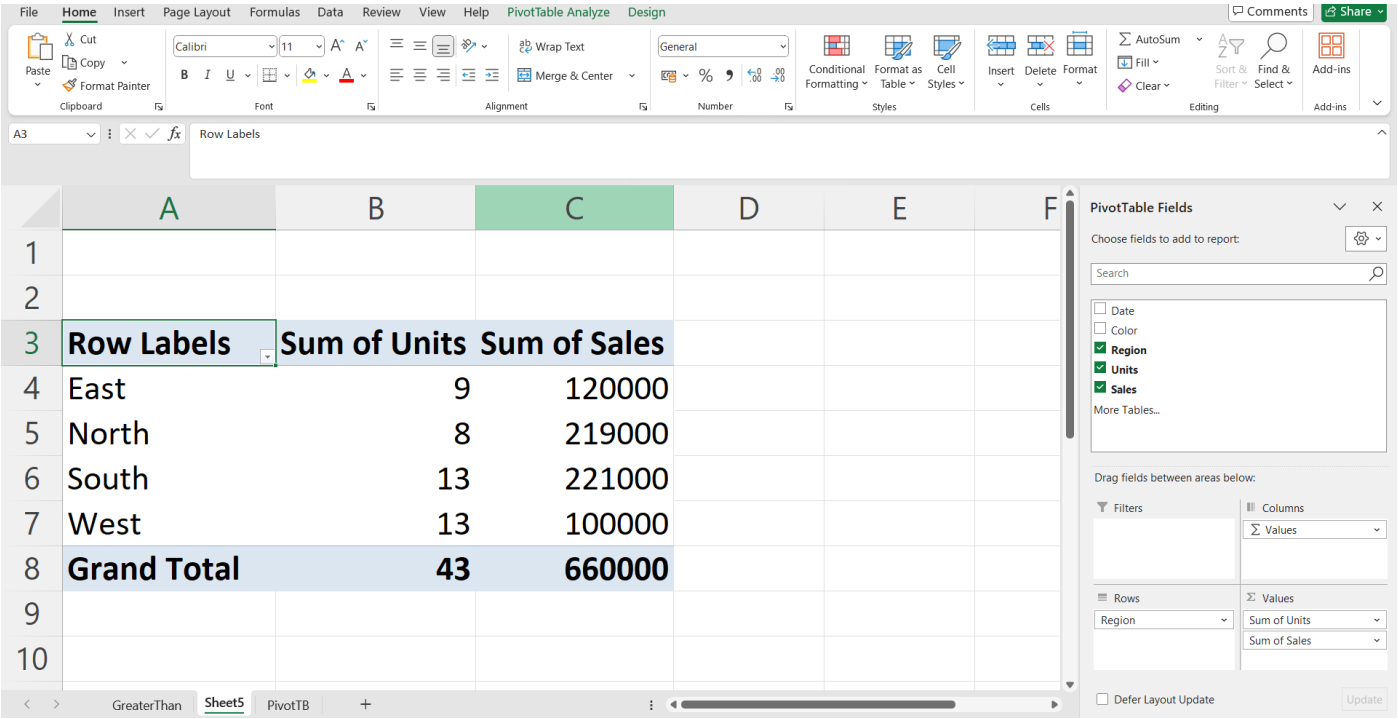


Date	Color	Region	Units	Sales
02-01-2026	Red	East	1	10000
03-01-2026	Green	West	5	15000
04-01-2026	Blue	North	4	169000
05-01-2026	Indigo	South	8	154000
06-01-2026	Violet	East	4	60000
07-01-2026	Yellow	West	2	40000
08-01-2026	Pink	North	4	50000
09-01-2026	Brown	South	5	67000
10-01-2026	Black	East	4	50000
11-01-2026	White	West	6	45000

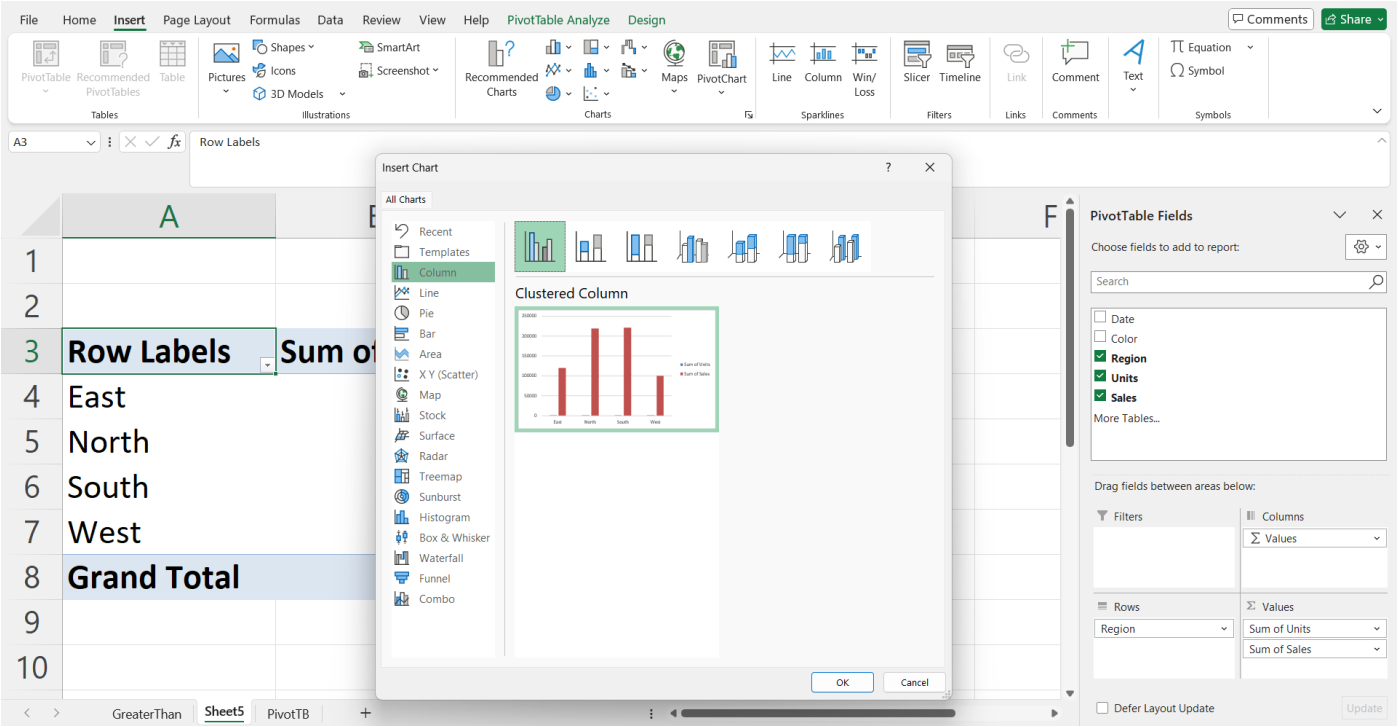
Step 2: Select the pivot chart and pivot table option from insert tab. Select "New worksheet" in the create pivot chart window. Select the table range and cell where you want the output.

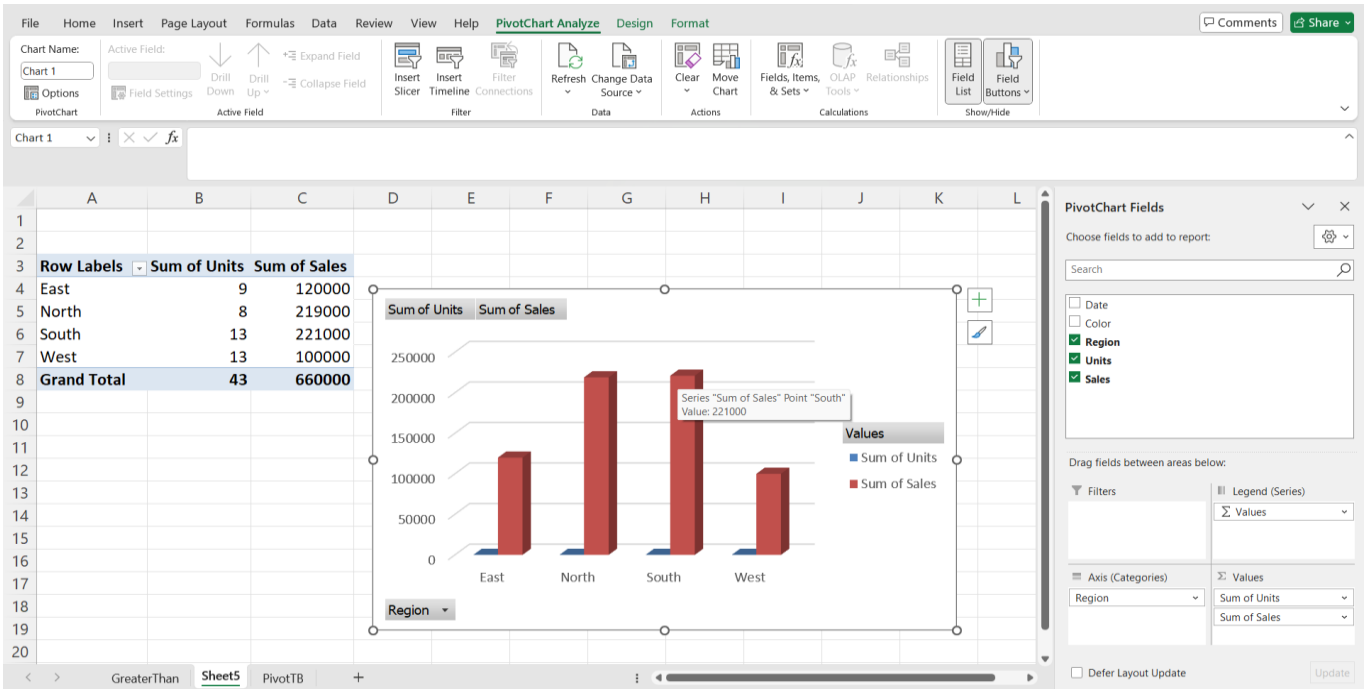


Step 3: Select and drag attributes in the below boxes.



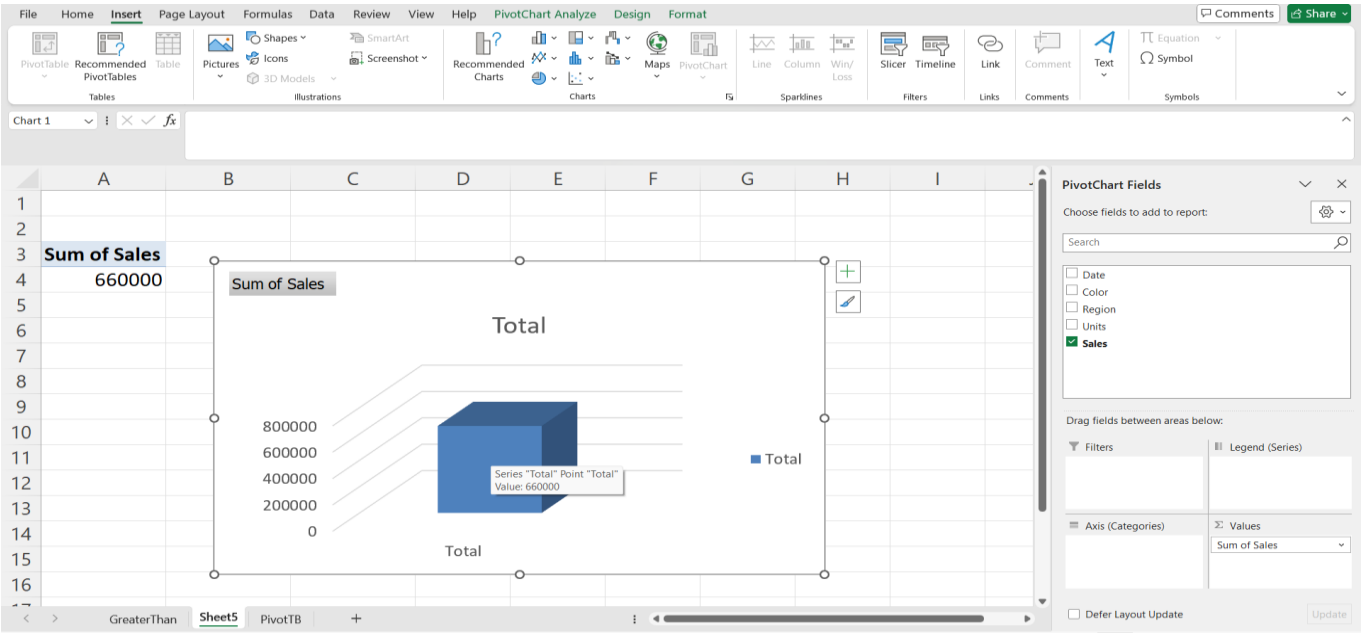
Step 4 : Select PivotChart from PivotTable Analyze Tab. Select the Chart Type and click ok.



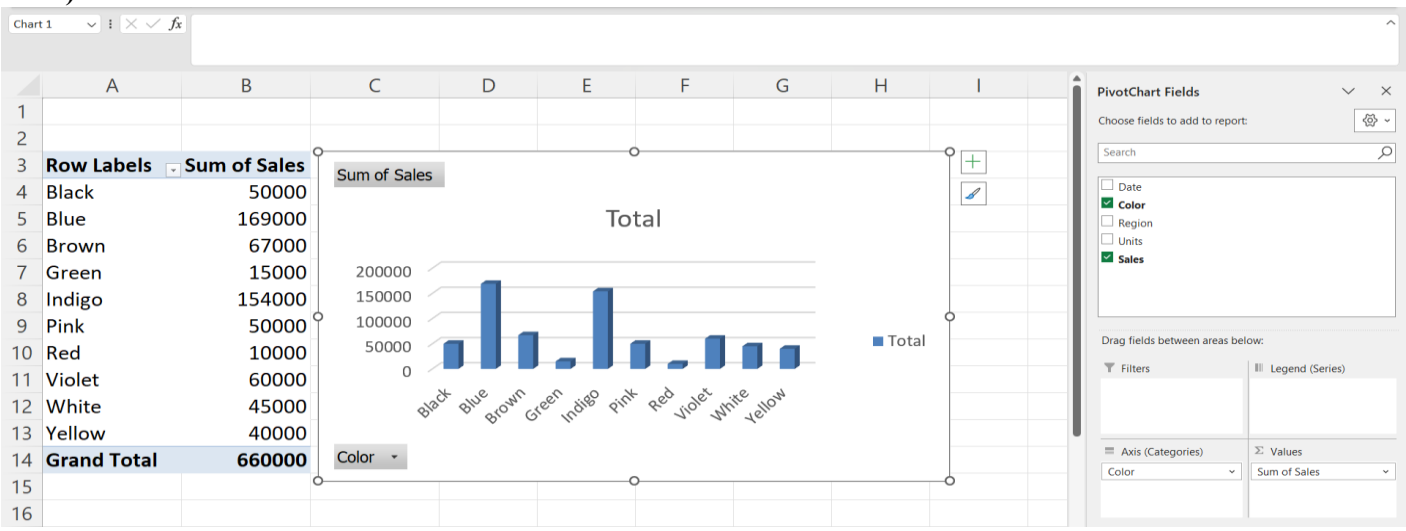


Step 5 : Perform the following Questions.

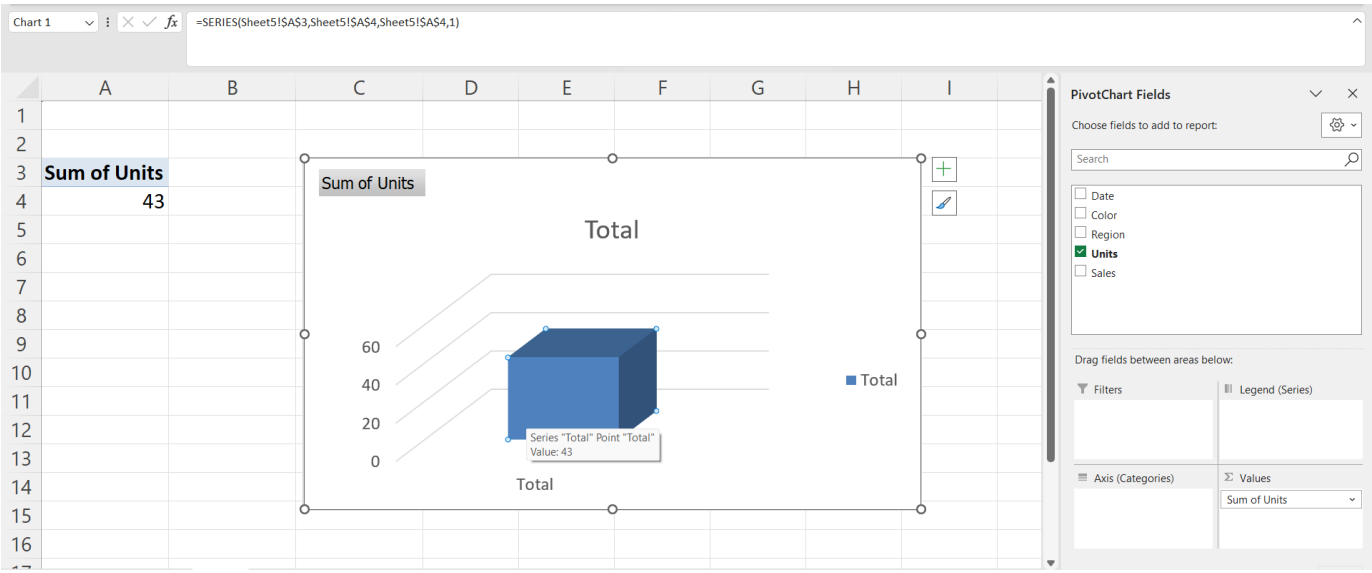
i) Find out the total sales.



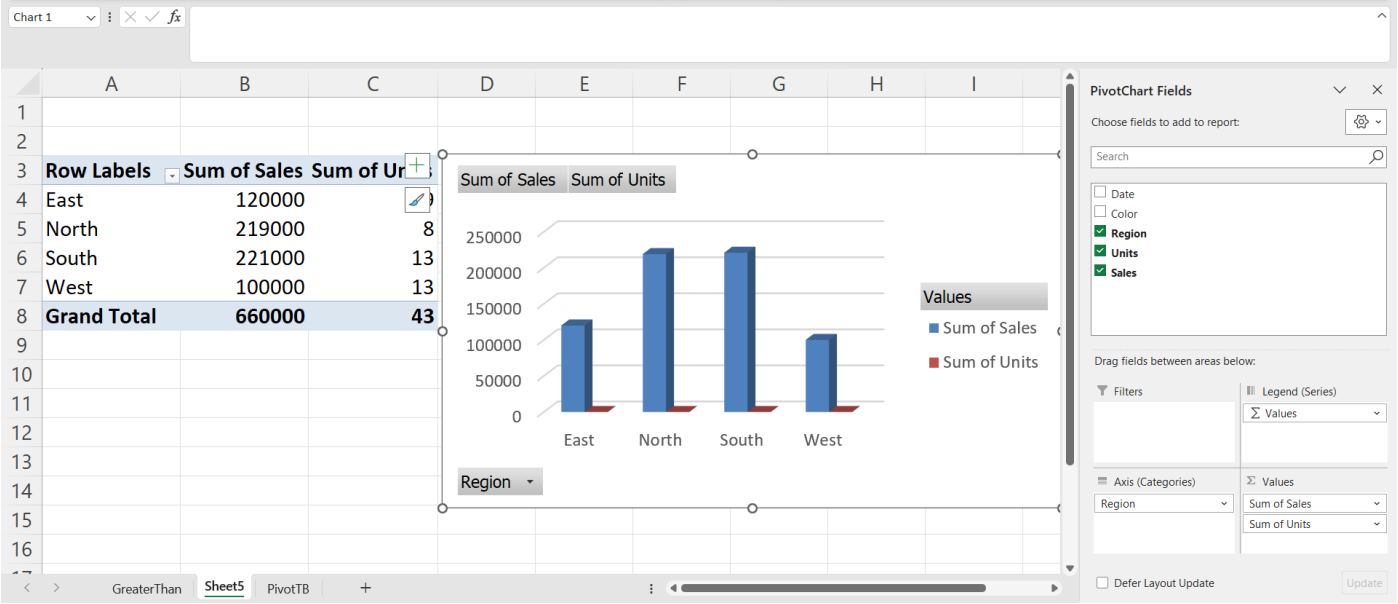
ii) Find out the sum of sales Color wise



iii) Find out the sum of units



iv) Find out Region wise total sales and total units



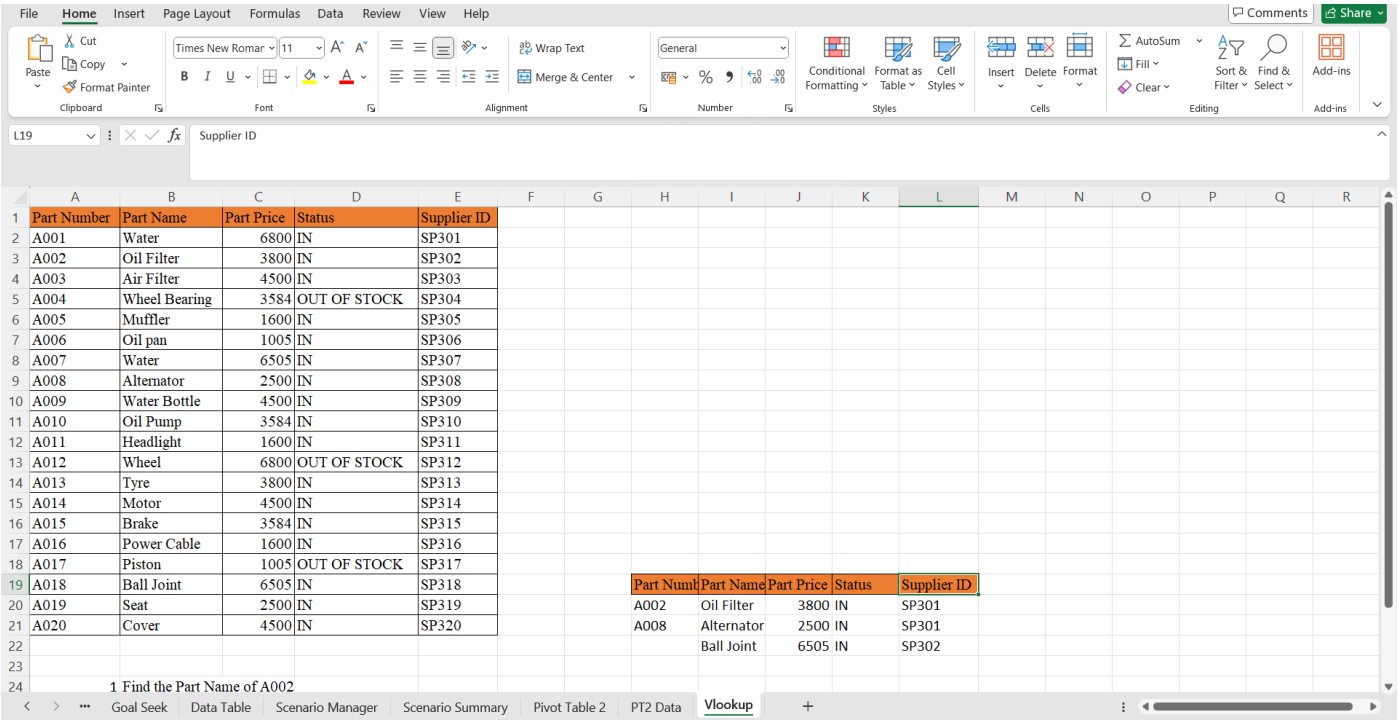
VLOOKUP

Formulas Tab → Function Library → Lookup & Reference → VLOOKUP

Formula : =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Steps

Step l : click on an empty cell and type the following command:
=VLOOKUP(A3,A2:E21,1,TRUE)



Part Number	Part Name	Part Price	Status	Supplier ID
A001	Water	6800	IN	SP301
A002	Oil Filter	3800	IN	SP302
A003	Air Filter	4500	IN	SP303
A004	Wheel Bearing	3584	OUT OF STOCK	SP304
A005	Muffler	1600	IN	SP305
A006	Oil pan	1005	IN	SP306
A007	Water	6505	IN	SP307
A008	Alternator	2500	IN	SP308
A009	Water Bottle	4500	IN	SP309
A010	Oil Pump	3584	IN	SP310
A011	Headlight	1600	IN	SP311
A012	Wheel	6800	OUT OF STOCK	SP312
A013	Tyre	3800	IN	SP313
A014	Motor	4500	IN	SP314
A015	Brake	3584	IN	SP315
A016	Power Cable	1600	IN	SP316
A017	Piston	1005	OUT OF STOCK	SP317
A018	Ball Joint	6505	IN	SP318
A019	Seat	2500	IN	SP319
A020	Cover	4500	IN	SP320

Part Numt	Part Name	Part Price	Status	Supplier ID
A002	Oil Filter	3800	IN	SP301
A008	Alternator	2500	IN	SP301
	Ball Joint	6505	IN	SP302

1 Find the Part Name of A002

i) Find the part name of part number “A002”
=VLOOKUP("A002", A2:E21, 2, FALSE)

ii) Find the Supplier ID for part name “Ball Joint”
=VLOOKUP("Ball Joint", B2:E21, 4, FALSE)

iii) Find the Status of Part number “A008”
=VLOOKUP("A008", A2:E21, 4, FALSE)

iv) Find the Part price for part name “Muffler”
=VLOOKUP("Muffler", B2:E21, 2, FALSE)

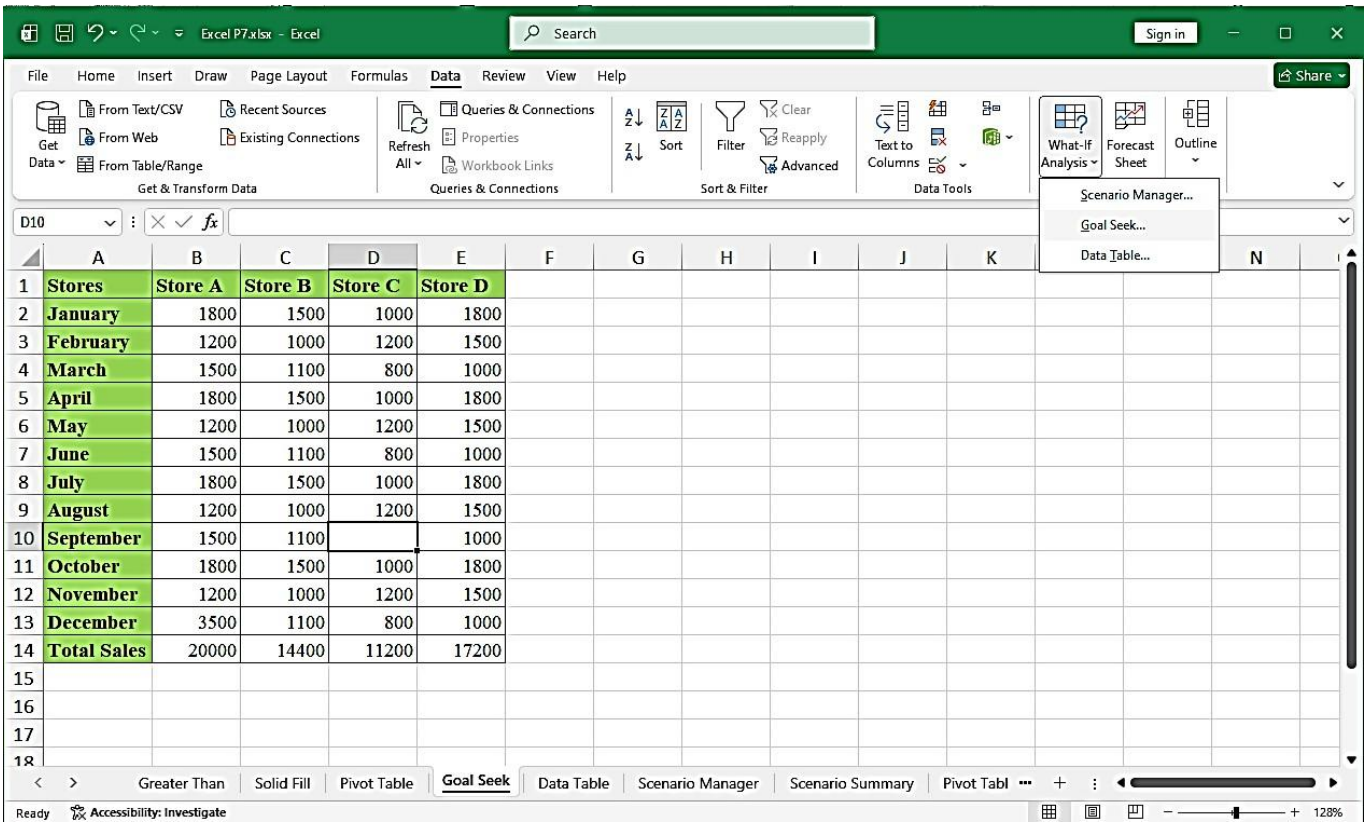
1	Find the Part Name of A002	Oil Filter
2	Find the supplier id for Ball Joint	SP318
3	Find the status of part number A008	IN
4	Find out the part price for part name muffler	1600

What-if-Analysis using Goal Seek

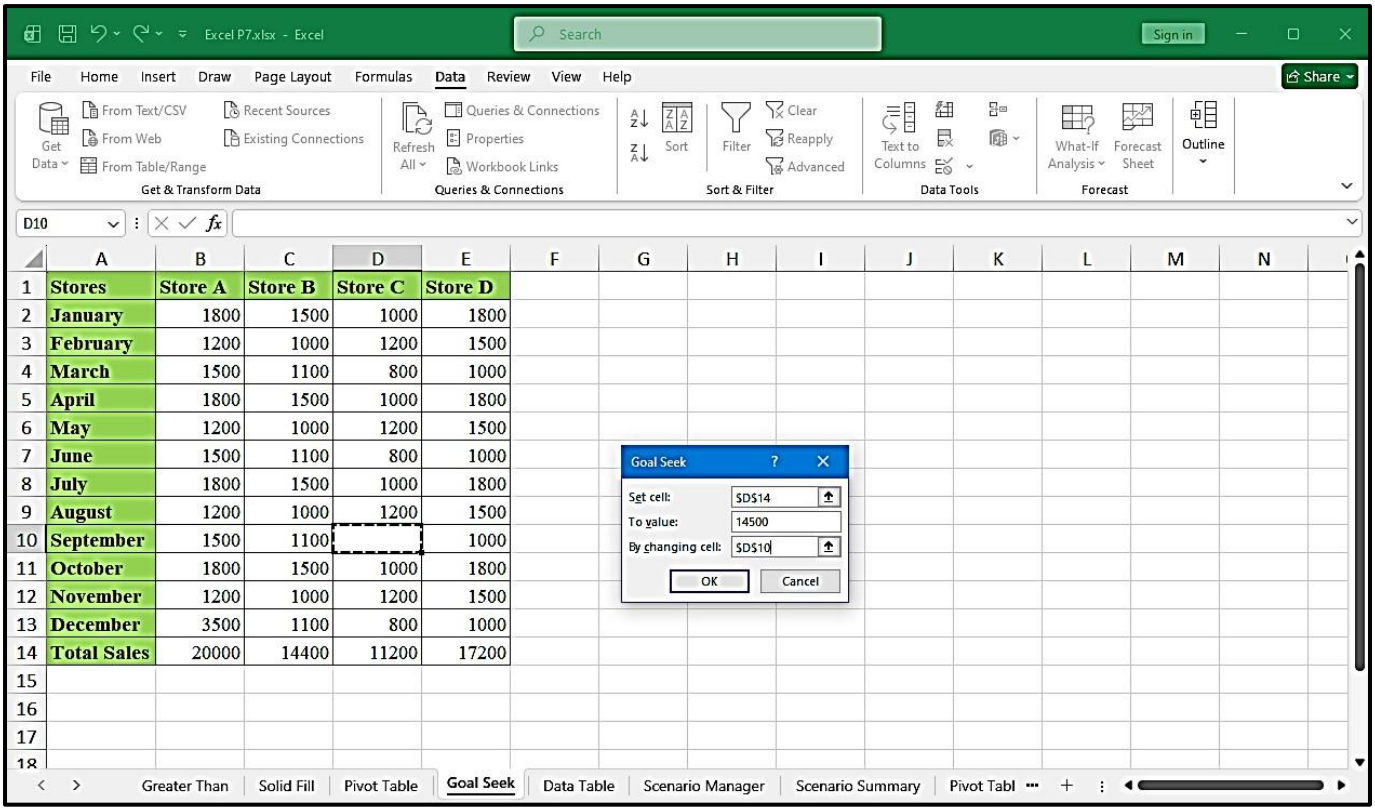
Data Tab → Forecast Group → What-If Analysis → Goal Seek

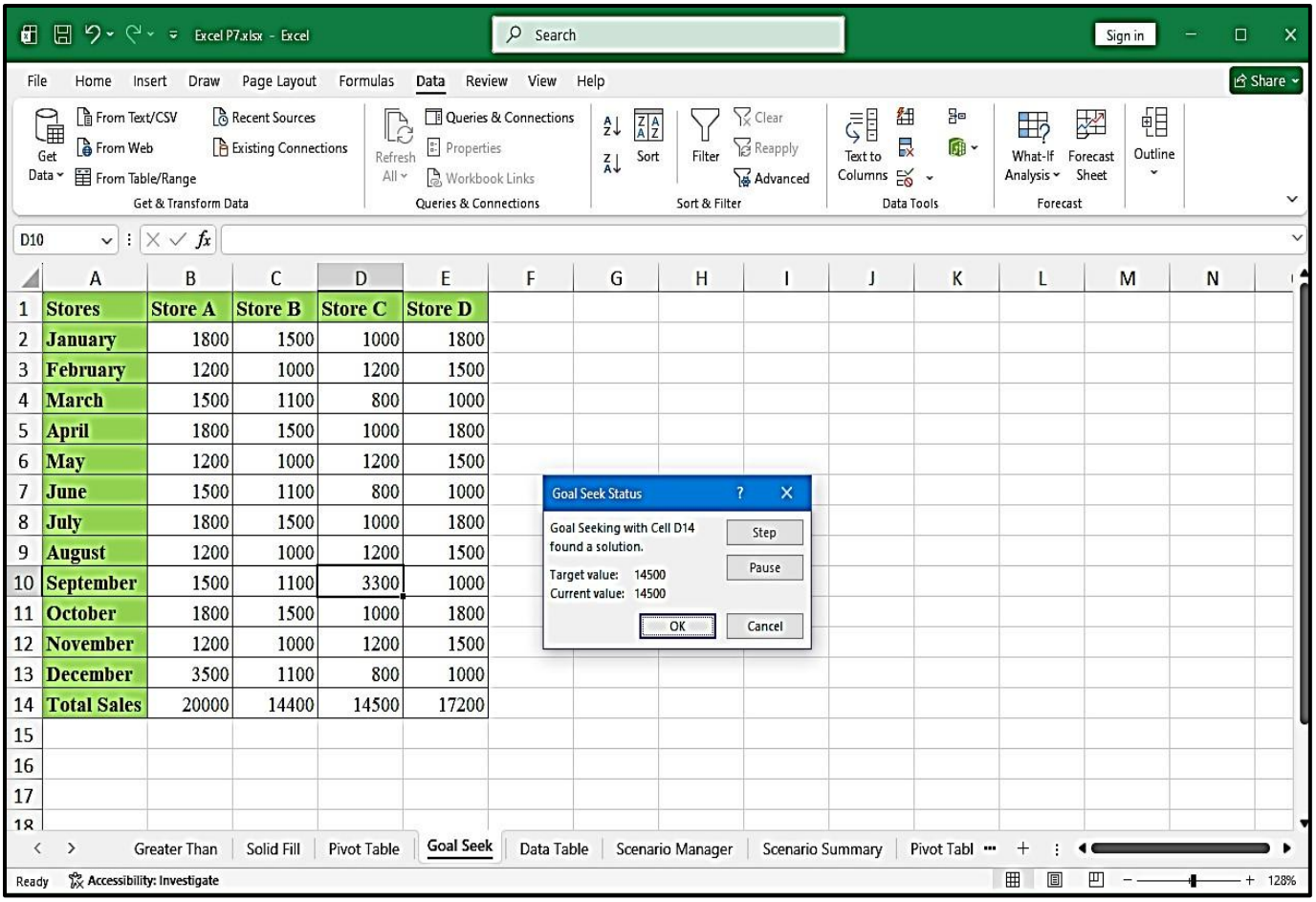
Steps

Step 1: In the Data tab go to the what if analysis>Goal seek.



Step 2: Fill the information in the window accordingly and click ok.





	A	B	C	D	E
1	Stores	Store A	Store B	Store C	Store D
2	January	1800	1500	1000	1800
3	February	1200	1000	1200	1500
4	March	1500	1100	800	1000
5	April	1800	1500	1000	1800
6	May	1200	1000	1200	1500
7	June	1500	1100	800	1000
8	July	1800	1500	1000	1800
9	August	1200	1000	1200	1500
10	September	1500	1100	3300	1000
11	October	1800	1500	1000	1800
12	November	1200	1000	1200	1500
13	December	3500	1100	800	1000
14	Total Sales	20000	14400	14500	17200

Practical 2

Aim: Data Frames and Basic Data Pre-processing

• Read data from CSV and JSON files into a data frame.

• Perform basic data pre-processing tasks such as handling missing values and outliers.

• Manipulate and transform data using functions like filtering, sorting, and grouping.

```
import pandas as pd
import numpy as np
# Reading a CSV file into a DataFrame
df=pd.read_csv(r"Iris.csv")
print(df.head()) # Display the first 5 rows of the DataFrame
```

```
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
```

```
# Step 2: Basic Data Exploration
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
df.tail()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```python
df.shape
```

```
(150, 6)
```

```python
df.dtypes
```

```
Id                 int64
SepalLengthCm    float64
SepalWidthCm     float64
PetalLengthCm    float64
PetalWidthCm     float64
Species           object
dtype: object
```

```python
df.columns
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
```

```python
# Step 3: Checking for Missing Values
# Checking for missing values in each column of the CSV DataFrame
missing_values = df.isnull().sum()
print("\nMissing Values in CSV Data:")
print(missing_values)
```

```
Missing Values in CSV Data:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

```python
df = df.drop(columns=['Species'])
# Step 4: Handling Missing Values
# We will fill missing values in columns with the mean of the column
# (You could also drop missing rows or use other strategies depending on your need
fill= df.fillna(df.mean())
print("\nFilled Missing Values with Mean (CSV Data):")
print(df.head())
```

```
Filled Missing Values with Mean (CSV Data):
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
0   1            5.1           3.5            1.4           0.2
1   2            4.9           3.0            1.4           0.2
2   3            4.7           3.2            1.3           0.2
3   4            4.6           3.1            1.5           0.2
4   5            5.0           3.6            1.4           0.2
```

```python
# Step 5: Handling Outliers
# Here we will calculate Z-scores and remove rows where Z-score is greater than 3
z_scores = np.abs((fill - fill.mean()) / fill.std())
z_scores
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1.714797	0.897674	1.028611	1.336794	1.308593
1	1.691780	1.139200	0.124540	1.336794	1.308593
2	1.668762	1.380727	0.336720	1.393470	1.308593
3	1.645745	1.501490	0.106090	1.280118	1.308593
4	1.622728	1.018437	1.259242	1.336794	1.308593
...	...	...	...	...	...
145	1.622728	1.034539	0.124540	0.816888	1.443121
146	1.645745	0.551486	1.277692	0.703536	0.918985
147	1.668762	0.793012	0.124540	0.816888	1.050019
148	1.691780	0.430722	0.797981	0.930239	1.443121
149	1.714797	0.068433	0.124540	0.760212	0.787951

150 rows × 5 columns

```
# Remove rows where any Z-score is greater than 3 (outliers)
do = fill[(z_scores < 3).all(axis=1)]
print("\nData After Removing Outliers (CSV Data):")
print(do.head())
```

```
Data After Removing Outliers (CSV Data):
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
0   1            5.1           3.5            1.4           0.2
1   2            4.9           3.0            1.4           0.2
2   3            4.7           3.2            1.3           0.2
3   4            4.6           3.1            1.5           0.2
4   5            5.0           3.6            1.4           0.2
```

```
# Step 6: Filtering Data (Example: Select rows where a column value is greater tha
threshold_value = 3 # Example threshold value
filter = do[do['SepalLengthCm'] > threshold_value]
print(f"\nFiltered Data (Rows with column_name > {threshold_value}):")
print(filter)
```

```
Filtered Data (Rows with column_name > 3):
      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
0      1            5.1           3.5            1.4           0.2
1      2            4.9           3.0            1.4           0.2
2      3            4.7           3.2            1.3           0.2
3      4            4.6           3.1            1.5           0.2
4      5            5.0           3.6            1.4           0.2
..   ...            ...           ...            ...           ...
145  146            6.7           3.0            5.2           2.3
146  147            6.3           2.5            5.0           1.9
147  148            6.5           3.0            5.2           2.0
148  149            6.2           3.4            5.4           2.3
149  150            5.9           3.0            5.1           1.8

[149 rows x 5 columns]
```

```python
# Step 7: Sorting Data (Sorting by a column in descending order)
df_sorted = filter.sort_values(by='SepalWidthCm', ascending=False)
print(df_sorted.head())
```

```
    Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
33  34            5.5           4.2            1.4           0.2
32  33            5.2           4.1            1.5           0.1
14  15            5.8           4.0            1.2           0.2
5    6            5.4           3.9            1.7           0.4
16  17            5.4           3.9            1.3           0.4
```

```python
# Step 8: Grouping Data (Example: Group by a column and calculate the mean of anot
df_grouped = df_sorted.groupby('SepalLengthCm').agg({
 'PetalLengthCm': 'mean', # Calculate the mean of 'another_column' for each gr
 'PetalWidthCm': 'sum' # Calculate the sum of 'yet_another_column' for each gr
}).reset_index() # Reset index to avoid multi-index
print("\nGrouped Data (Mean and Sum for Each Group):")
print(df_grouped.head())
```

```
Grouped Data (Mean and Sum for Each Group):
   SepalLengthCm  PetalLengthCm  PetalWidthCm
0            4.3       1.100000           0.1
1            4.4       1.333333           0.6
2            4.5       1.300000           0.3
3            4.6       1.325000           0.9
4            4.7       1.450000           0.4
```

Practical 3

Aim: Feature Scaling and Dummification

• Apply feature-scaling techniques like standardization and normalization to numerical features.

• Perform feature dummification to convert categorical variables into numerical representations.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

df = pd.read_csv("wine.csv")
print(df)
```

```
     Wine  Alcohol  Malic.acid   Ash   Acl   Mg  Phenols  Flavanoids  \
0       1    14.23        1.71  2.43  15.6  127     2.80        3.06
1       1    13.20        1.78  2.14  11.2  100     2.65        2.76
2       1    13.16        2.36  2.67  18.6  101     2.80        3.24
3       1    14.37        1.95  2.50  16.8  113     3.85        3.49
4       1    13.24        2.59  2.87  21.0  118     2.80        2.69
..    ...      ...         ...   ...   ...  ...      ...         ...
173     3    13.71        5.65  2.45  20.5   95     1.68        0.61
174     3    13.40        3.91  2.48  23.0  102     1.80        0.75
175     3    13.27        4.28  2.26  20.0  120     1.59        0.69
176     3    13.17        2.59  2.37  20.0  120     1.65        0.68
177     3    14.13        4.10  2.74  24.5   96     2.05        0.76

     Nonflavanoid.phenols  Proanth  Color.int   Hue    OD  Proline
0                    0.28     2.29       5.64  1.04  3.92     1065
1                    0.26     1.28       4.38  1.05  3.40     1050
2                    0.30     2.81       5.68  1.03  3.17     1185
3                    0.24     2.18       7.80  0.86  3.45     1480
4                    0.39     1.82       4.32  1.04  2.93      735
..                    ...      ...        ...   ...   ...      ...
173                  0.52     1.06       7.70  0.64  1.74      740
174                  0.43     1.41       7.30  0.70  1.56      750
175                  0.43     1.35      10.20  0.59  1.56      835
176                  0.53     1.46       9.30  0.60  1.62      840
177                  0.56     1.35       9.20  0.61  1.60      560

[178 rows x 14 columns]
```

```
df1 = pd.read_csv("wine.csv", usecols=[0,1,2], skiprows=1)
df1.columns = ['classlabel1', 'Alcohol', 'Malic Acid']
print(df1)
```

```
     classlabel1  Alcohol  Malic Acid
0              1    13.20        1.78
1              1    13.16        2.36
2              1    14.37        1.95
3              1    13.24        2.59
4              1    14.20        1.76
..           ...      ...         ...
172            3    13.71        5.65
173            3    13.40        3.91
174            3    13.27        4.28
175            3    13.17        2.59
176            3    14.13        4.10

[177 rows x 3 columns]
```

```
scaling = MinMaxScaler()
Scaled_value = scaling.fit_transform(df1[['Alcohol', 'Malic Acid']])
df1[['Alcohol', 'Malic Acid']] = Scaled_value
print(df1)
```

```
     classlabel1   Alcohol  Malic Acid
0              1  0.571053    0.205534
1              1  0.560526    0.320158
2              1  0.878947    0.239130
3              1  0.581579    0.365613
4              1  0.834211    0.201581
..           ...       ...         ...
172            3  0.705263    0.970356
173            3  0.623684    0.626482
174            3  0.589474    0.699605
175            3  0.563158    0.365613
176            3  0.815789    0.664032

[177 rows x 3 columns]
```

```
scaling = StandardScaler()
scaled_standardvalue = scaling.fit_transform(df1[['Alcohol', 'Malic Acid']])
df1[['Alcohol', 'Malic Acid']] = scaled_standardvalue
print(df1)
```

```
     classlabel1   Alcohol  Malic Acid
0              1  0.255824   -0.501624
1              1  0.206229    0.018020
2              1  1.706501   -0.349315
3              1  0.305420    0.224086
4              1  1.495719   -0.519543
..           ...       ...         ...
172            3  0.888171    2.965658
173            3  0.503803    1.406725
174            3  0.342617    1.738222
175            3  0.218628    0.224086
176            3  1.408926    1.576953

[177 rows x 3 columns]
```

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder
iris = pd.read_csv('Iris.csv')
print(iris)
```

```
      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  \
0      1            5.1           3.5            1.4           0.2
1      2            4.9           3.0            1.4           0.2
2      3            4.7           3.2            1.3           0.2
3      4            4.6           3.1            1.5           0.2
4      5            5.0           3.6            1.4           0.2
..   ...            ...           ...            ...           ...
145  146            6.7           3.0            5.2           2.3
146  147            6.3           2.5            5.0           1.9
147  148            6.5           3.0            5.2           2.0
148  149            6.2           3.4            5.4           2.3
149  150            5.9           3.0            5.1           1.8

            Species
0       Iris-setosa
1       Iris-setosa
2       Iris-setosa
3       Iris-setosa
4       Iris-setosa
..              ...
145  Iris-virginica
146  Iris-virginica
147  Iris-virginica
148  Iris-virginica
149  Iris-virginica

[150 rows x 6 columns]
```

```python
le = LabelEncoder()
iris['code'] = le.fit_transform(iris.Species)
print(iris)
```

```
      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  \
0      1            5.1           3.5            1.4           0.2
1      2            4.9           3.0            1.4           0.2
2      3            4.7           3.2            1.3           0.2
3      4            4.6           3.1            1.5           0.2
4      5            5.0           3.6            1.4           0.2
..   ...            ...           ...            ...           ...
145  146            6.7           3.0            5.2           2.3
146  147            6.3           2.5            5.0           1.9
147  148            6.5           3.0            5.2           2.0
148  149            6.2           3.4            5.4           2.3
149  150            5.9           3.0            5.1           1.8

            Species  code
0       Iris-setosa     0
1       Iris-setosa     0
2       Iris-setosa     0
3       Iris-setosa     0
4       Iris-setosa     0
..              ...   ...
145  Iris-virginica     2
146  Iris-virginica     2
147  Iris-virginica     2
148  Iris-virginica     2
149  Iris-virginica     2

[150 rows x 7 columns]
```

Practical 4

Aim: Hypothesis Testing

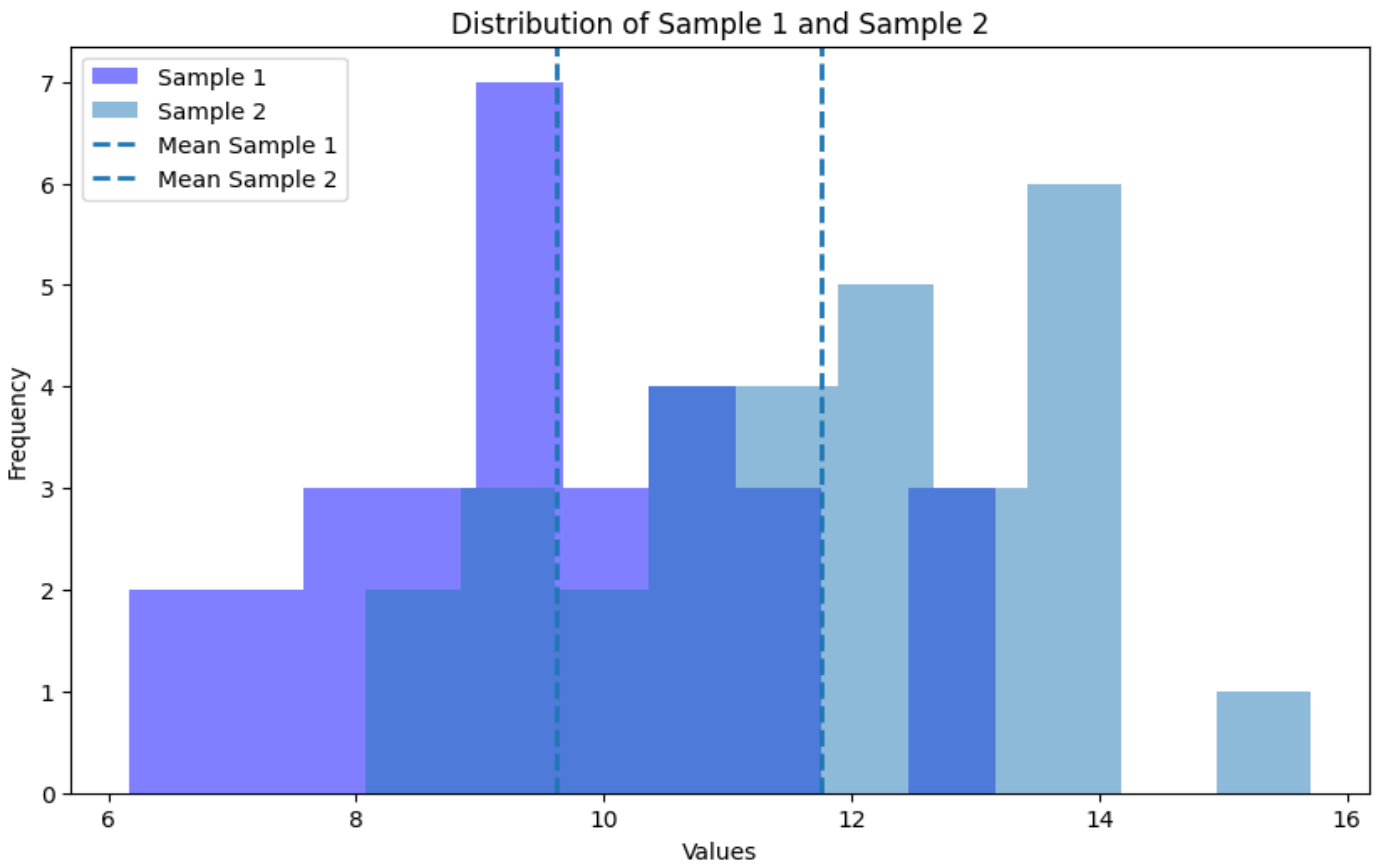
• Formulate null and alternative hypotheses for a given problem.

• Conduct a hypothesis test using appropriate statistical tests (e.g., t-test, chi-square test).

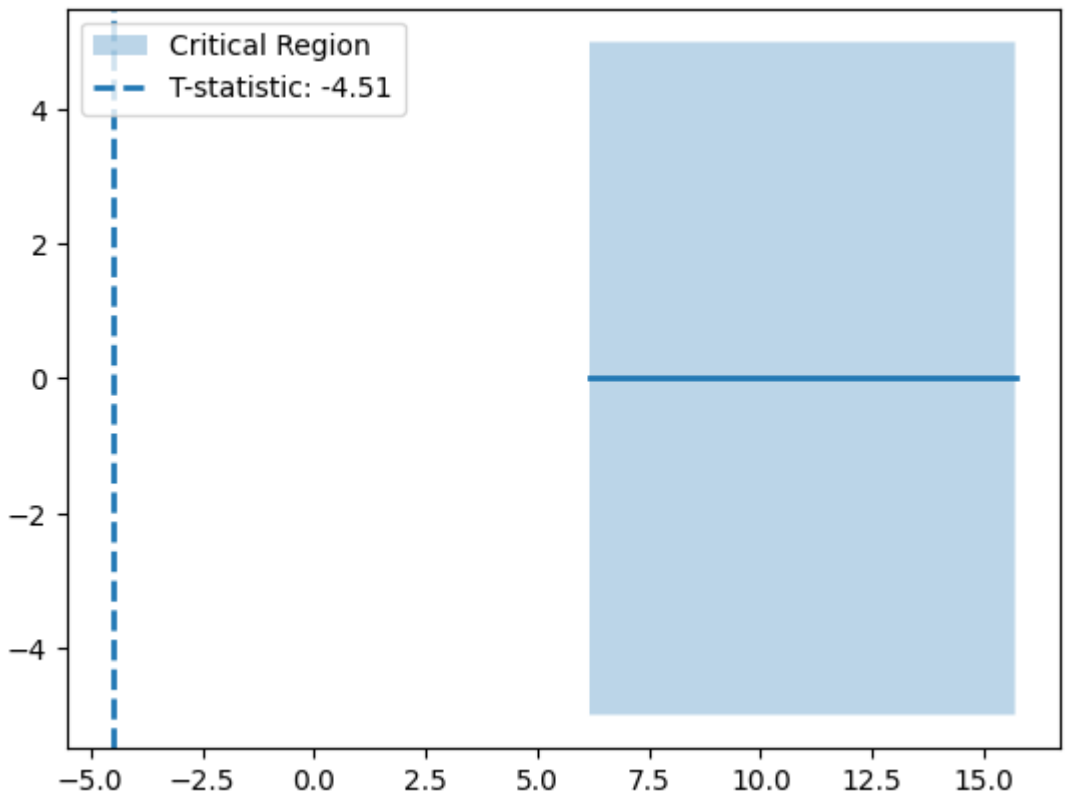
• Interpret the results and draw conclusions based on the test outcomes.

```
# t-test
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Generate two samples for demonstrate purposes
np.random.seed(42)
sample1 = np.random.normal(loc=10, scale=2, size=30)
sample2 = np.random.normal(loc=12, scale=2, size=30)
# Perform a two-sample t-test
t_statistic, p_value = stats.ttest_ind(sample1, sample2)

# Set the signifiacnt level
alpha = 0.05
print("Results of Two-sample t-test:")
print(f'T-statistic:{t_statistic}')
print(f'P-value:{p_value}')
print(f"Degree of Freedom: {len(sample1) + len(sample2)-2}")

# Plot the distributions
plt.figure(figsize = (10, 6))
plt.hist(sample1, alpha = 0.5, label = 'Sample 1', color = 'blue')
plt.hist(sample2, alpha=0.5, label='Sample 2')

plt.axvline(np.mean(sample1), linestyle='--', linewidth=2, label='Mean Sample 1')
plt.axvline(np.mean(sample2), linestyle='--', linewidth=2, label='Mean Sample 2')

plt.title('Distribution of Sample 1 and Sample 2')
plt.xlabel('Values')
plt.ylabel('Frequency')
plt.legend()

plt.show()
```

```
Results of Two-sample t-test:
T-statistic:-4.512913234547555
P-value:3.176506547470154e-05
Degree of Freedom: 58
```




```python
# Highlight the critical region if null hypothesis is rejected
if p_value < alpha:
    critical_region = np.linspace(
        min(sample1.min(), sample2.min()),
        max(sample1.max(), sample2.max()),
        1000
    )

    # Draw a baseline instead of an empty canvas
    plt.plot(critical_region, np.zeros_like(critical_region), linewidth=2)

    # Highlight critical region as a narrow band
    plt.fill_between(
        critical_region,
        -5,
        5,
        alpha=0.3,
        label='Critical Region'
    )

    # Mark the t-statistic clearly
    plt.axvline(
        t_statistic,
        linestyle='--',
        linewidth=2,
        label=f"T-statistic: {t_statistic:.2f}"
    )

    plt.legend()
    plt.show()
```




```
# Draw Conclusions
if p_value < alpha:
    if np.mean(sample1) > np.mean(sample2):
        print("Conclusion: There is significant evidence to reject the null hypothesis.")
        print("Interpretation: The mean of Sample 1 is significantly higher than that of
Sample 2.")
    else:
        print("Conclusion: There is significant evidence to reject the null hypothesis.")
        print("Interpretation: The mean of Sample 2 is significantly higher than that of
Sample 1.")
else:
    print("Conclusion: Fail to reject the null hypothesis.")
    print("Interpretation: There is not enough evidence to claim a significant difference
between the means.")
```

```
Conclusion: There is significant evidence to reject the null hypothesis.
Interpretation: The mean of Sample 2 is significantly higher than that of Sample 1.
```

```
#chi-test

import pandas as pd
import numpy as np
import matplotlib as plt
import seaborn as sb
import warnings

from scipy import stats

warnings.filterwarnings('ignore')
```

```python
df=sb.load_dataset('mpg')
print(df)

print(df['horsepower'].describe())
print(df['model_year'].describe())

bins=[0,75,150,240]
df['horsepower_new']=pd.cut(df['horsepower'],bins=bins,labels=['l','m','h'])
c=df['horsepower_new']
print(c)

ybins=[69,72,74,84]
label=['t1','t2','t3']
df['modelyear_new']=pd.cut(df['model_year'],bins=ybins,labels=label)
newyear=df['modelyear_new']
print(newyear)

df_chi=pd.crosstab(df['horsepower_new'],df['modelyear_new'])
print(df_chi)

print(stats.chi2_contingency(df_chi))
```

```
      mpg  cylinders  displacement  horsepower  weight  acceleration  \
0    18.0          8         307.0       130.0    3504          12.0
1    15.0          8         350.0       165.0    3693          11.5
2    18.0          8         318.0       150.0    3436          11.0
3    16.0          8         304.0       150.0    3433          12.0
4    17.0          8         302.0       140.0    3449          10.5
..    ...        ...           ...         ...     ...           ...
393  27.0          4         140.0        86.0    2790          15.6
394  44.0          4          97.0        52.0    2130          24.6
395  32.0          4         135.0        84.0    2295          11.6
396  28.0          4         120.0        79.0    2625          18.6
397  31.0          4         119.0        82.0    2720          19.4

     model_year  origin                       name
0            70     usa  chevrolet chevelle malibu
1            70     usa          buick skylark 320
2            70     usa         plymouth satellite
3            70     usa              amc rebel sst
4            70     usa                ford torino
..          ...     ...                        ...
393          82     usa            ford mustang gl
394          82  europe                  vw pickup
395          82     usa         dodge rampage
396          82     usa                ford ranger
397          82     usa                 chevy s-10

[398 rows x 9 columns]
count    392.000000
mean     104.469388
std       38.491160
min       46.000000
25%       75.000000
50%       93.500000
75%      126.000000
max      230.000000
Name: horsepower, dtype: float64
count    398.000000
mean      76.010050
std        3.697627
min       70.000000
```

```
25%       73.000000
50%       76.000000
75%       79.000000
max       82.000000
Name: model_year, dtype: float64
0      m
1      h
2      m
3      m
4      m
      ..
393    m
394    l
395    m
396    m
397    m
Name: horsepower_new, Length: 398, dtype: category
Categories (3, object): ['l' < 'm' < 'h']
0      t1
1      t1
2      t1
3      t1
4      t1
      ..
393    t3
394    t3
395    t3
396    t3
397    t3
Name: modelyear_new, Length: 398, dtype: category
Categories (3, object): ['t1' < 't2' < 't3']
modelyear_new   t1  t2   t3
horsepower_new
l                9  14   76
m               49  41  158
h               26  11    8
Chi2ContingencyResult(statistic=np.float64(54.95485392447537), pvalue=np.fl
       [ 53.14285714,  41.75510204, 153.10204082],
       [  9.64285714,   7.57653061,  27.78061224]]))
```

Practical 5

Aim: ANOVA (Analysis of Variance)

• Perform one-way ANOVA to compare means across multiple groups.

• Conduct post-hoc tests to identify significant differences between group means.

```
import pandas as pd
from scipy.stats import f_oneway
from statsmodels.stats.multicomp import pairwise_tukeyhsd

# Step 1: Load Dataset
df = pd.read_csv("Iris.csv")

print(df.head())
```

```
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
```

```
# Step 2: Separate Groups
# Using SepalLengthCm across Species
setosa = df[df['Species'] == 'Iris-setosa']['SepalLengthCm']
versicolor = df[df['Species'] == 'Iris-versicolor']['SepalLengthCm']
virginica = df[df['Species'] == 'Iris-virginica']['SepalLengthCm']

# Step 3: One-Way ANOVA
f_statistic, p_value = f_oneway(setosa, versicolor, virginica)

print("\nOne-Way ANOVA Result")
print("F-statistic:", f_statistic)
print("P-value:", p_value)

# Step 4: Decision
if p_value < 0.05:
    print("Result: Reject Null Hypothesis (Significant Difference Exists)")
else:
    print("Result: Fail to Reject Null Hypothesis (No Significant Difference)")

# Step 5: Post-Hoc Test (Tukey HSD)
tukey = pairwise_tukeyhsd(
    endog=df['SepalLengthCm'],
    groups=df['Species'],
    alpha=0.05
)

print("\nPost-Hoc Test (Tukey HSD) Results:")
print(tukey)
```

```
One-Way ANOVA Result
F-statistic: 119.26450218450468
P-value: 1.669669190769383e-31
Result: Reject Null Hypothesis (Significant Difference Exists)

Post-Hoc Test (Tukey HSD) Results:
        Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================================
     group1          group2     meandiff p-adj lower  upper  reject
---------------------------------------------------------------------
    Iris-setosa Iris-versicolor     0.93   0.0 0.6862 1.1738   True
    Iris-setosa  Iris-virginica    1.582   0.0 1.3382 1.8258   True
Iris-versicolor  Iris-virginica    0.652   0.0 0.4082 0.8958   True
---------------------------------------------------------------------
```

Practical 6

Aim: Regression and Its Types

• Implement simple linear regression using a dataset.

• Explore and interpret the regression model coefficients and goodness-of-fit measures.

• Extend the analysis to multiple linear regression and assess the impact of additional predictors.

```
import pandas as pd
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score
df = pd.read_csv(r"housing.csv")
df.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.45857	5.682861	7.009188	4.09	23086.80050	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.64245	6.002900	6.730821	3.09	40173.07217	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.06718	5.865890	8.512727	5.13	36882.15940	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3	63345.24005	7.188236	5.586729	3.26	34310.24283	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.19723	5.040555	7.839388	4.23	26354.10947	6.309435e+05	USNS Raymond\nFPO AE 09386

```
df.tail()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
4995	60567.94414	7.830362	6.137356	3.46	22837.36103	1060193.786	USNS Williams\nFPO AP 30153-7653
4996	78491.27543	6.999135	6.576763	4.02	25616.11549	1482617.729	PSC 9258, Box 8489\nAPO AA 42991-3352
4997	63390.68689	7.250591	4.805081	2.13	33266.14549	1030729.583	4215 Tracy Garden Suite 076\nJoshualand, VA 01...
4998	68001.33124	5.534388	7.130144	5.44	42625.62016	1198656.872	USS Wallace\nFPO AE 73316
4999	65510.58180	5.992305	6.792336	4.07	46501.28380	1298950.480	37778 George Ridges Apt. 509\nEast Holly, NV 2...

```
df.shape
```

```
(5000, 7)
```

```
df.size
```

```
35000
```

```
df.describe()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562390	5.322283	6.299250	3.140000	29403.928700	9.975771e+05
50%	68804.286405	5.970429	7.002902	4.050000	36199.406690	1.232669e+06
75%	75783.338665	6.650808	7.665871	4.490000	42861.290770	1.471210e+06
max	107701.748400	9.519088	10.759588	6.500000	69621.713380	2.469066e+06

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

```python
df.dtypes
```

```
Avg. Area Income                float64
Avg. Area House Age             float64
Avg. Area Number of Rooms       float64
Avg. Area Number of Bedrooms    float64
Area Population                 float64
Price                           float64
Address                          object
dtype: object
```

```python
#import ssl
#ssl._create_default_https_context = ssl._create_unverified_context
housing = fetch_california_housing()
# Convert to DataFrame
housing_df = pd.DataFrame(housing.data, columns=housing.feature_names)
housing_df.head() # Print first few rows
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25

```python
housing_df['PRICE']=housing.target

X=housing_df[['AveRooms']]

y=housing_df[['PRICE']]
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=LinearRegression()
model.fit(X_train,y_train)
```

LinearRegression

▶ Parameters

```
mse=mean_squared_error(y_test,model.predict(X_test))

r2=r2_score(y_test,model.predict(X_test))
print("Mean Squared Error: ", mse)
print("R-squared: ",r2)
print("Intercept: ",model.intercept_)
print("Co-efficient: ",model.coef_)
```

```
Mean Squared Error:  1.2923314440807299
R-squared:  0.013795337532284901
Intercept:  [1.65476227]
Co-efficient:  [[0.07675559]]
```

```
X = housing_df.drop('PRICE',axis=1)

y = housing_df['PRICE']
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
model = LinearRegression()
model.fit(X_train,y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test,y_pred)
r2 = r2_score(y_test,y_pred)
print("Mean Squared Error:",mse)
print("R-squared:",r2)
print("Intercept:",model.intercept_)
print("Coefficient:",model.coef_)
```

```
Mean Squared Error: 0.555891598695244
R-squared: 0.5757877060324511
Intercept: -37.023277706064064
Coefficient: [ 4.48674910e-01  9.72425752e-03 -1.23323343e-01  7.83144907e-01
 -2.02962058e-06 -3.52631849e-03 -4.19792487e-01 -4.33708065e-01]
```

Practical 7

Aim: Logistic Regression and Decision Tree

• Build a logistic regression model to predict a binary outcome.

• Evaluate the model's performance using classification metrics (e.g., accuracy, precision, recall).

• Construct a decision tree model and interpret the decision rules for classification

```
import numpy as numpy
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

#Load dataset
df = pd.read_csv("Iris.csv")
df = df[df['Species'] != 'Iris-virginica']

X = df.drop('Species', axis =1)
y = df['Species']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size = 0.2, random_state = 42
)

logistic_model = LogisticRegression()
logistic_model.fit(X_train, y_train)

# Predictions
y_pred_logistic = logistic_model.predict(X_test)

# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred_logistic))
print("\nClassification Report")
print(classification_report(y_test, y_pred_logistic))
```

```
Accuracy: 1.0

Classification Report
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        12
Iris-versicolor       1.00      1.00      1.00         8

       accuracy                           1.00        20
      macro avg       1.00      1.00      1.00        20
   weighted avg       1.00      1.00      1.00        20
```

```python
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Create Model
model = DecisionTreeClassifier()

# Train Model
model.fit(X_train, y_train)

# Predictions
y_pred_tree = model.predict(X_test)

# Evaluation
print("\nDecision Tree Metrics")
print("Accuracy: ", accuracy_score(y_test, y_pred_tree))
print("\nClassification Report")
print(classification_report(y_test, y_pred_tree))
```

```
Decision Tree Metrics
Accuracy:  1.0

Classification Report
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        12
Iris-versicolor       1.00      1.00      1.00         8

       accuracy                           1.00        20
      macro avg       1.00      1.00      1.00        20
   weighted avg       1.00      1.00      1.00        20
```

Practical 8

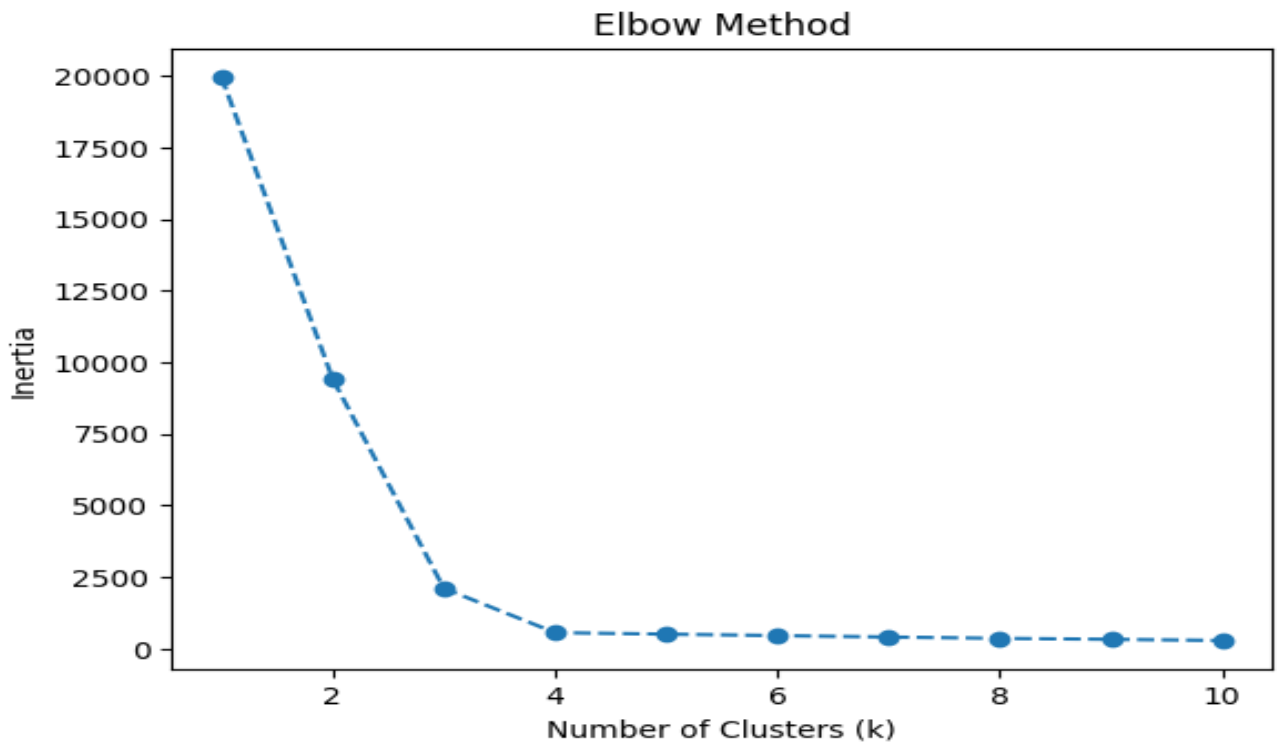
Aim: K-Means Clustering

• Apply the K-Means algorithm to group similar data points into clusters.

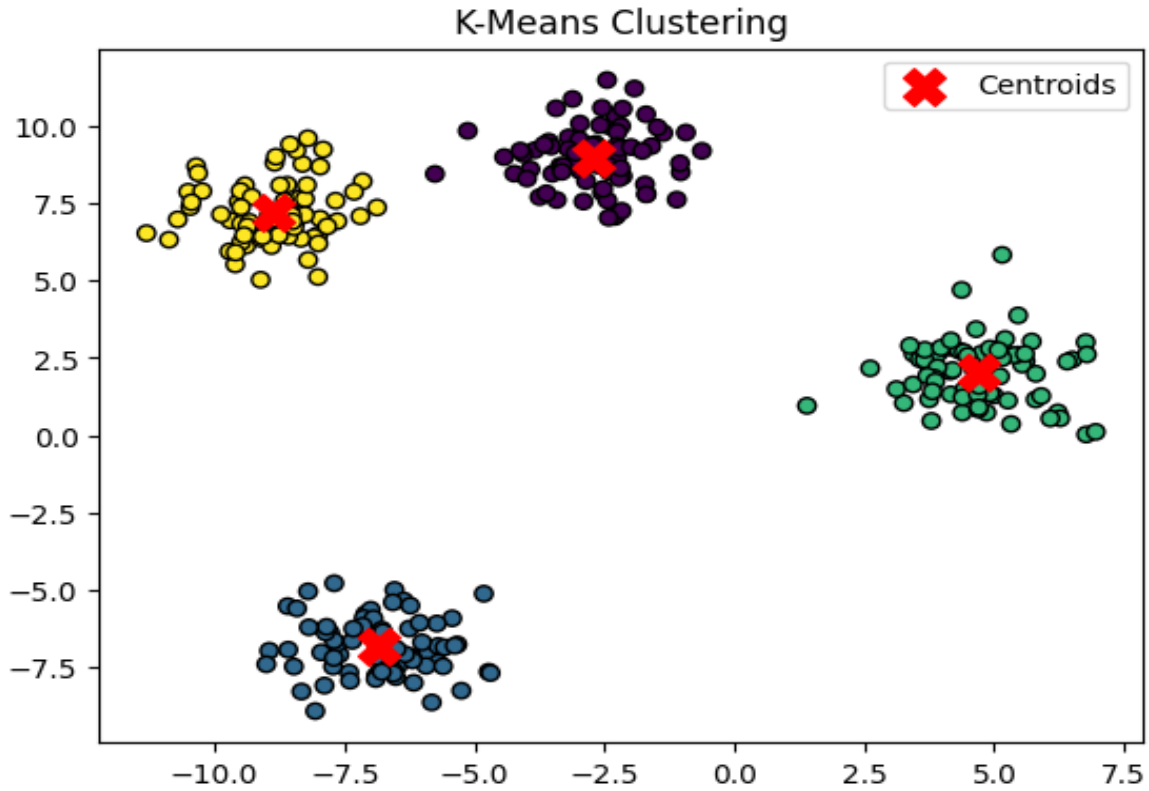
• Determine the optimal number of clusters using elbow method or silhouette analysis.

• Visualize the clustering results and analyze the cluster characteristics.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs


# Generate synthetic data
X, _ = make_blobs(n_samples=300, centers=4, random_state=42)
# Step 1: Elbow Method to find the optimal number of clusters
inertia = []
K_range = range(1, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X)
    inertia.append(kmeans.inertia_)
K_range = range(1, 11)
inertia = []

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X)
    inertia.append(kmeans.inertia_)

plt.plot(K_range, inertia, marker='o', linestyle='--')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()
```




```
# Step 2: Apply K-Means with the chosen k (let's pick k=4)
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X)


# Step 3: Visualize Clustering Results
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, cmap='viridis', edgecolors='k')
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
 s=200, c='red', marker='X', label='Centroids')
plt.title('K-Means Clustering')
plt.legend()
plt.show()
```



Practical 9

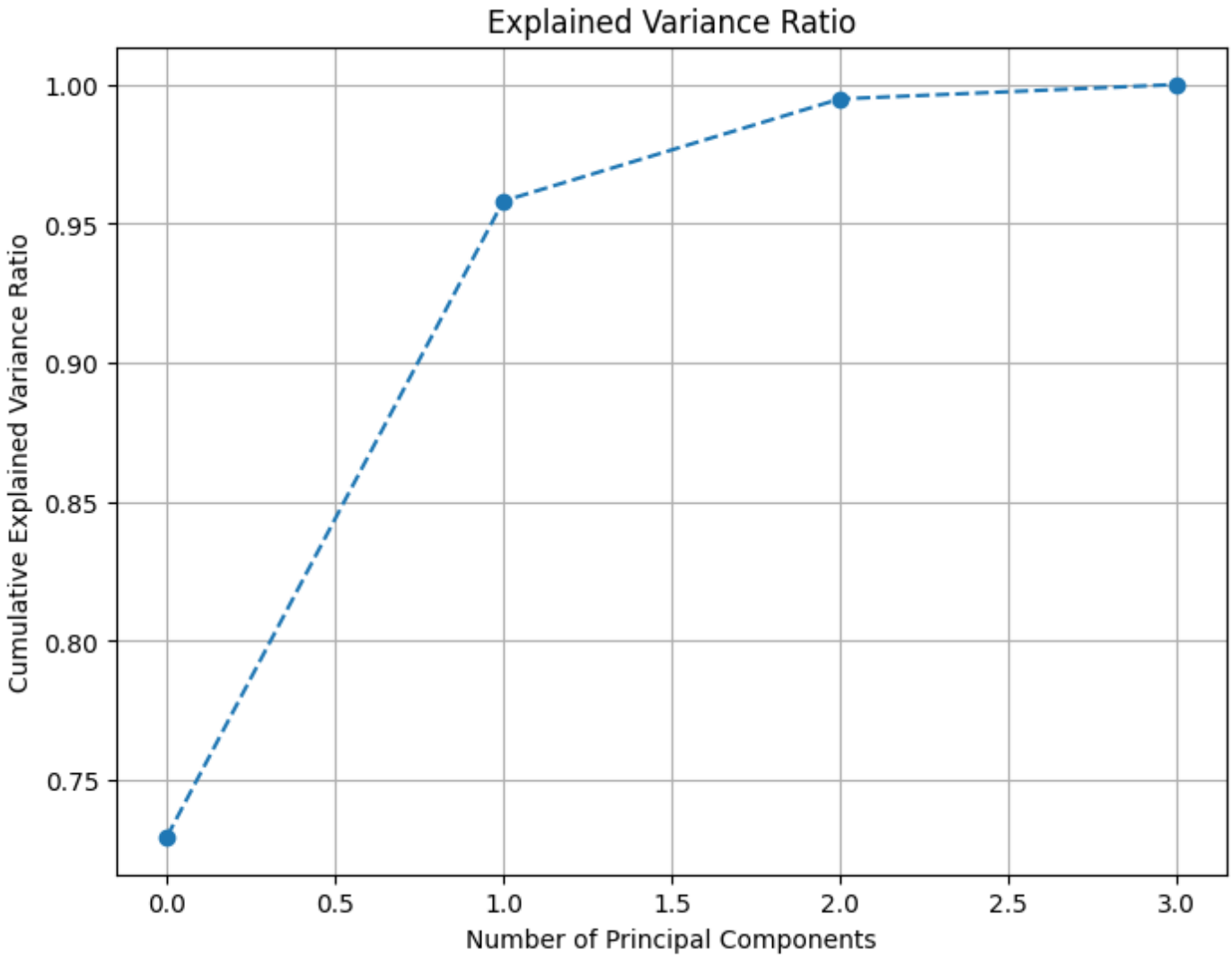
Aim: Principal Component Analysis (PCA)

• Perform PCA on a dataset to reduce dimensionality.

• Evaluate the explained variance and select the appropriate number of principal components.

• Visualize the data in the reduced-dimensional space.

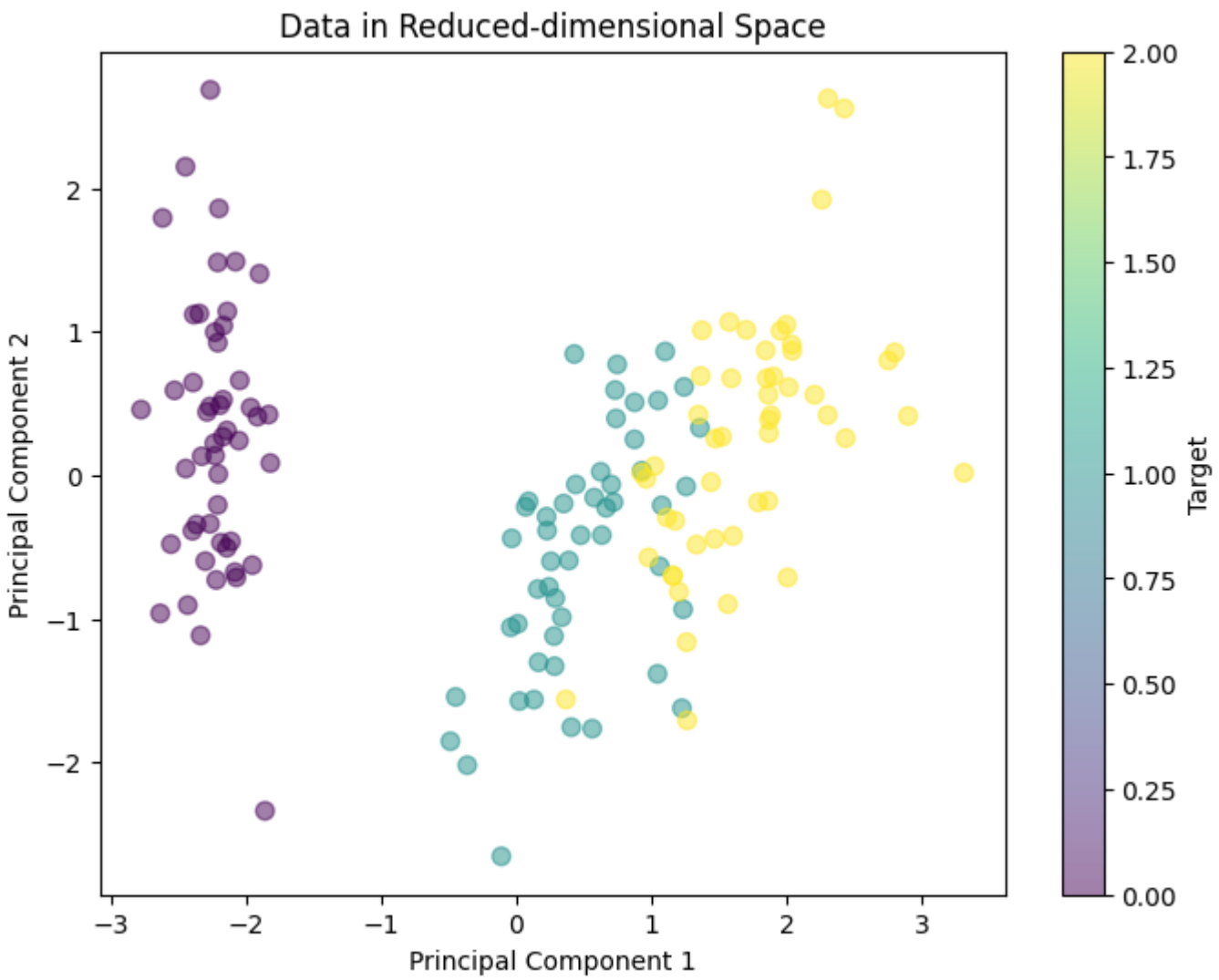
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load the Iris dataset
iris = load_iris()

iris_df = pd.DataFrame(
    data=np.c_[iris['data'], iris['target']],
    columns=iris['feature_names'] + ['target']
)

# Features and target
X = iris_df.drop('target', axis=1)
y = iris_df['target']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
pca = PCA()                 # <-- pca is defined HERE
X_pca = pca.fit_transform(X_scaled)

explained_variance_ratio = pca.explained_variance_ratio_
plt.figure(figsize=(8, 6))
plt.plot(
    np.cumsum(explained_variance_ratio),
    marker='o',
    linestyle='--'
)
plt.title('Explained Variance Ratio')
plt.xlabel('Number of Principal Components')
plt.ylabel('Cumulative Explained Variance Ratio')
plt.grid(True)
plt.show()
```




```
cumulative_variance_ratio = np.cumsum(explained_variance_ratio)
n_components = np.argmax(cumulative_variance_ratio >= 0.95) + 1

print(f'Number of principal components to explain 95% variance: {n_components}')
```

```
Number of principal components to explain 95% variance: 2
```

```python
pca = PCA(n_components=n_components)
X_reduced = pca.fit_transform(X_scaled)
plt.figure(figsize=(8, 6))
plt.scatter(
    X_reduced[:, 0],
    X_reduced[:, 1],
    c=y,
    cmap='viridis',
    s=50,
    alpha=0.5
)
plt.title('Data in Reduced-dimensional Space')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.colorbar(label='Target')
plt.show()
```



Practical 10

Aim: Data Visualization and Storytelling

• Create meaningful visualizations using data visualization tools.

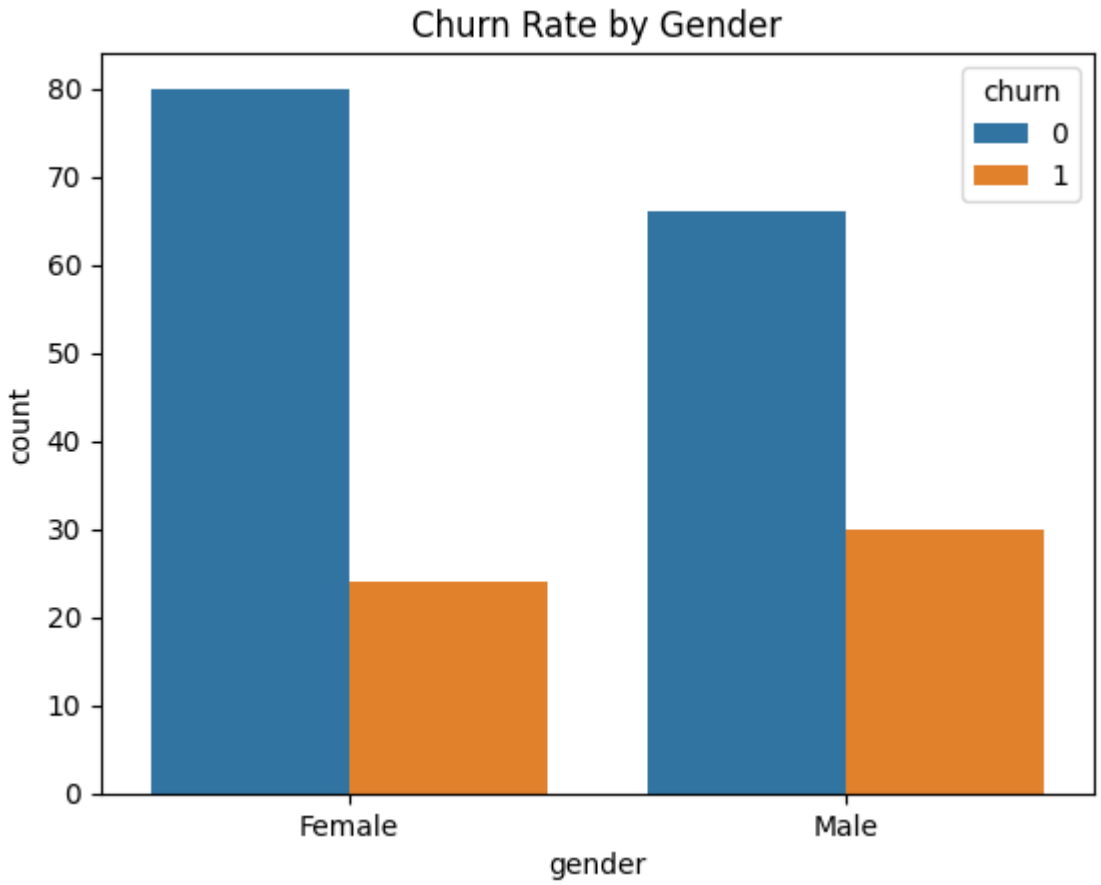
• Combine multiple visualizations to tell a compelling data story.

• Present the findings and insights in a clear and concise manner.

```
%pip install seaborn
%pip install plotly
%pip install nbformat
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
df = pd.read_csv(r"customer_data.csv")
df.head()
```

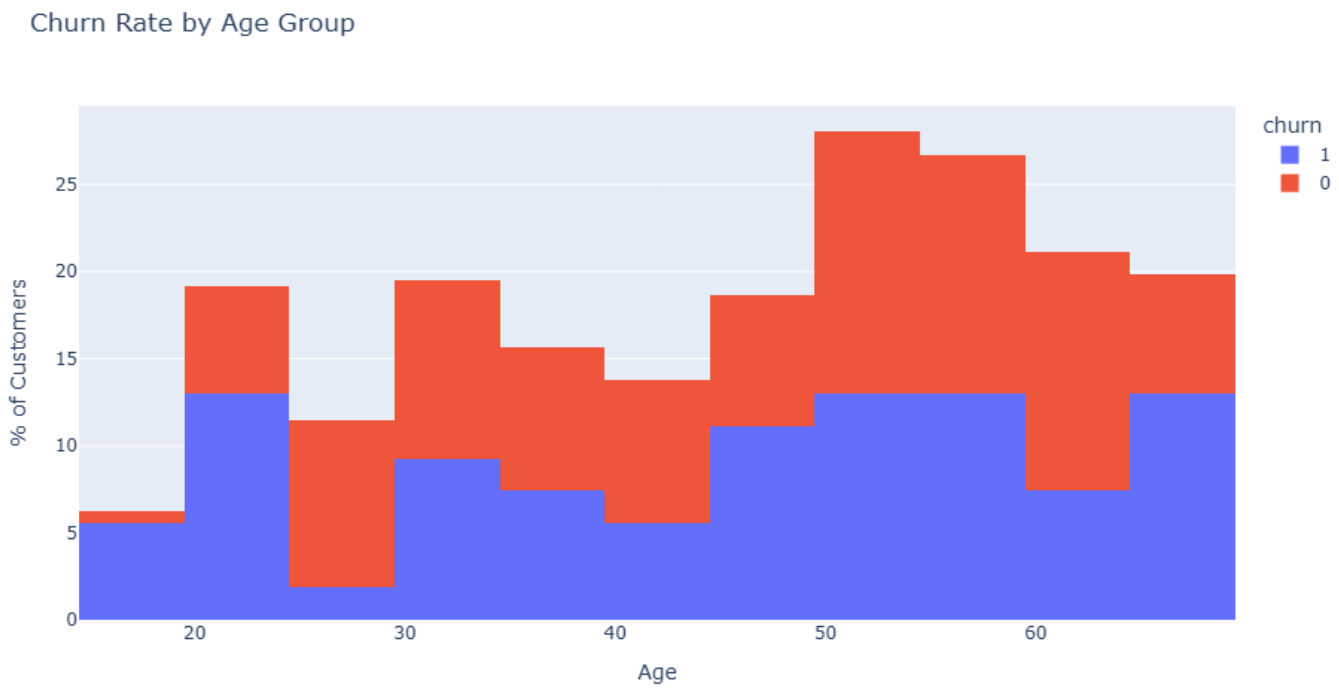
	customer_id	gender	age	service_type	tenure	churn
0	1	Female	22	VIP	3	1
1	2	Male	48	Premium	34	1
2	3	Male	50	VIP	59	0
3	4	Male	35	Basic	50	0
4	5	Male	42	Basic	52	0

```
sns.countplot(data=df, x='gender', hue='churn')
plt.title('Churn Rate by Gender')
plt.show()
gender_churn = df.groupby('gender')['churn'].mean()
print("\nChurn rate by gender:")
print(gender_churn)
```

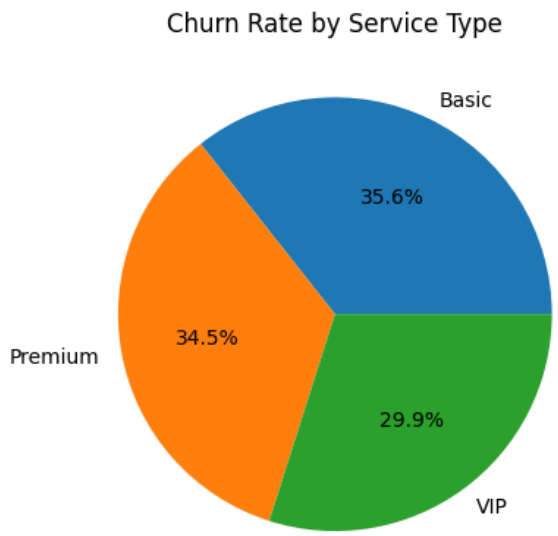


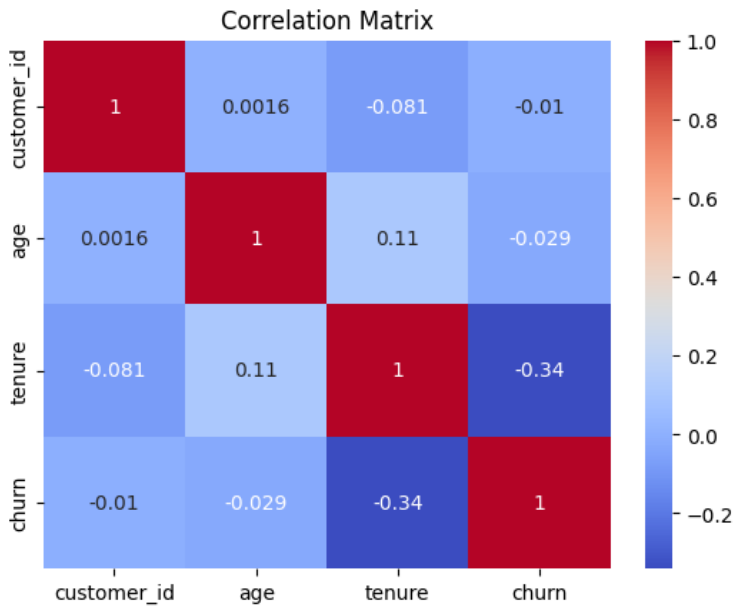

```
Churn rate by gender:
gender
Female    0.230769
Male      0.312500
Name: churn, dtype: float64
```

```
import plotly.io as pio
# Use 'iframe' for JupyterLite for the nbformat check
pio.renderers.default = 'iframe'
import plotly.io as pio
import plotly.express as px

# Fix for the nbformat error
pio.renderers.default = "notebook_connected"

fig = px.histogram(
    df,
    x='age',
    color='churn',
    nbins=20,
    histnorm='percent',
    title='Churn Rate by Age Group'
)
fig.update_layout(xaxis_title='Age', yaxis_title='% of Customers')
fig.show()
```

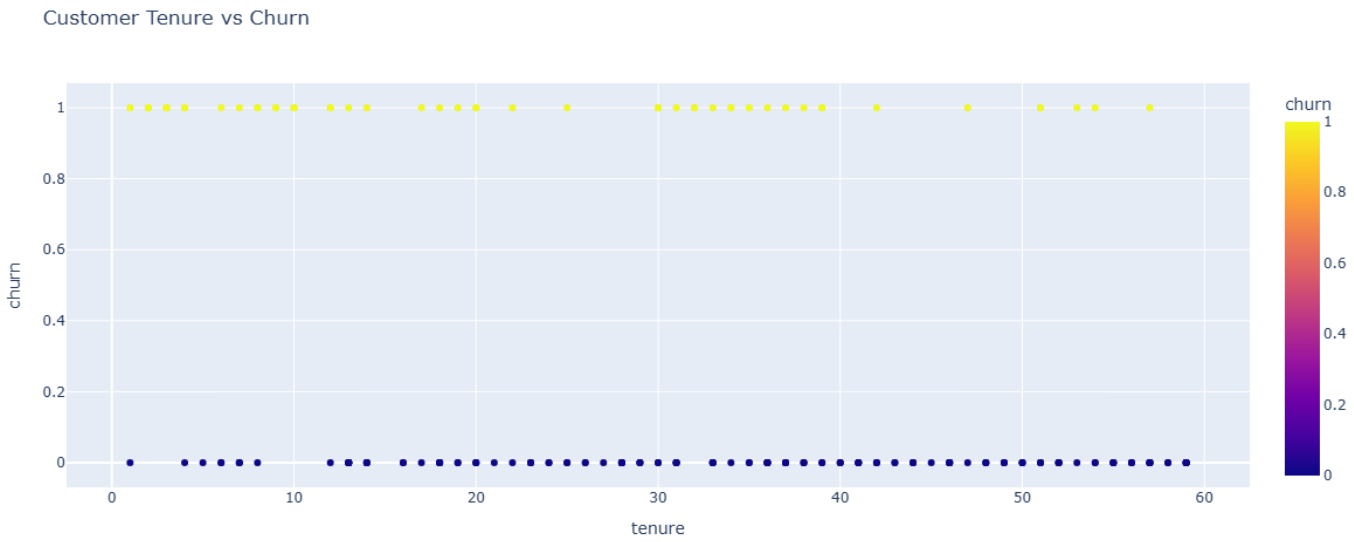



```
service_churn = df.groupby('service_type')['churn'].mean()
plt.pie(service_churn, labels=service_churn.index, autopct='%1.1f%%')
plt.title('Churn Rate by Service Type')
plt.show()

correlation_matrix = df.corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```





```python
fig = px.scatter(df, x='tenure', y='churn', color='churn', title='Customer Tenure vs
Churn')
fig.show()

print("\nKey Insights from Data Visualization...")
print(f"1. Gender: Churn is {df.groupby('gender')['churn'].mean().to_dict()}")
print(f"2. Service: Highest churn in
{df.groupby('service_type')['churn'].mean().idxmax()}")
print(f"3. Tenure: Lower tenure correlates strongly with higher churn.")
print(f"4. Top Factors:
{df.corr(numeric_only=True)['churn'].sort_values(ascending=False).index[1:4].tolist()}")
```




```
Key Insights from Data Visualization...
1. Gender: Churn is {'Female': 0.23076923076923078, 'Male': 0.3125}
2. Service: Highest churn in Basic
3. Tenure: Lower tenure correlates strongly with higher churn.
4. Top Factors: ['customer_id', 'age', 'tenure']
```